

Pinellas County Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

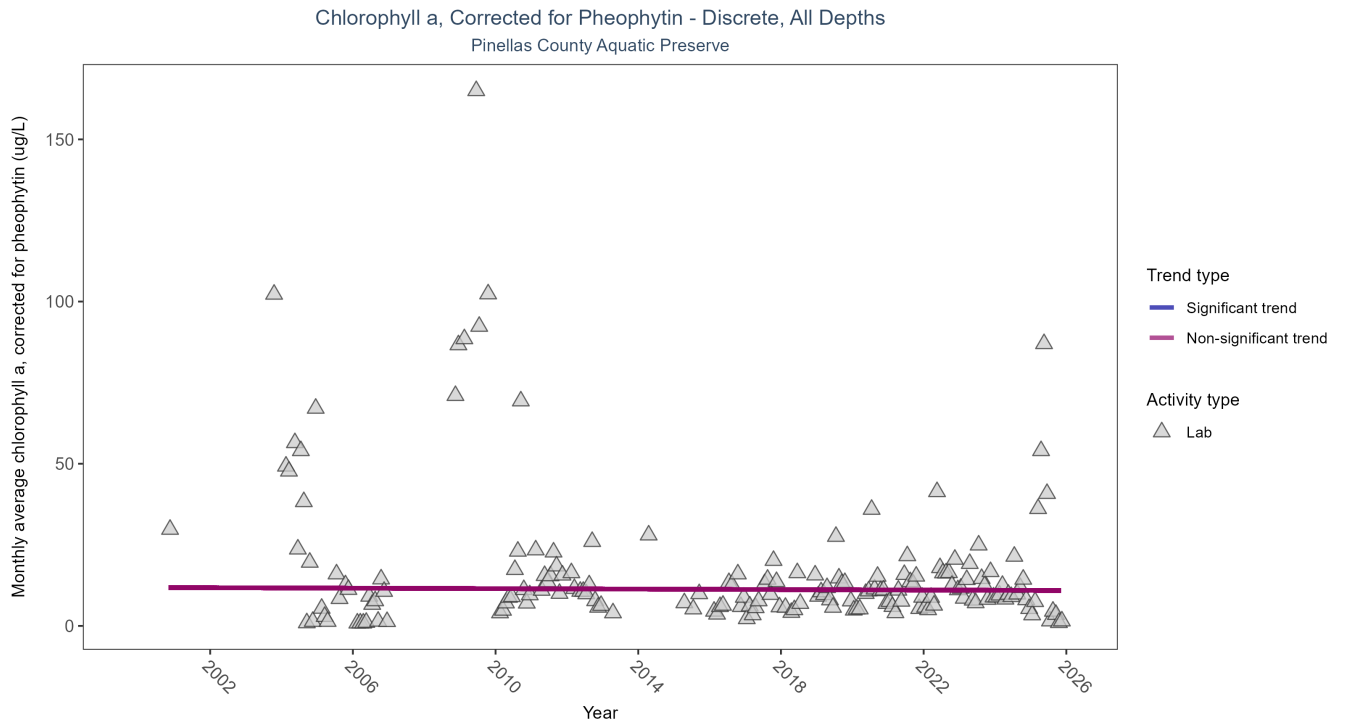


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3684	23	2000 - 2025	5.8	-0.008	11.8486	-0.0359	0.779

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2000 and 2025.

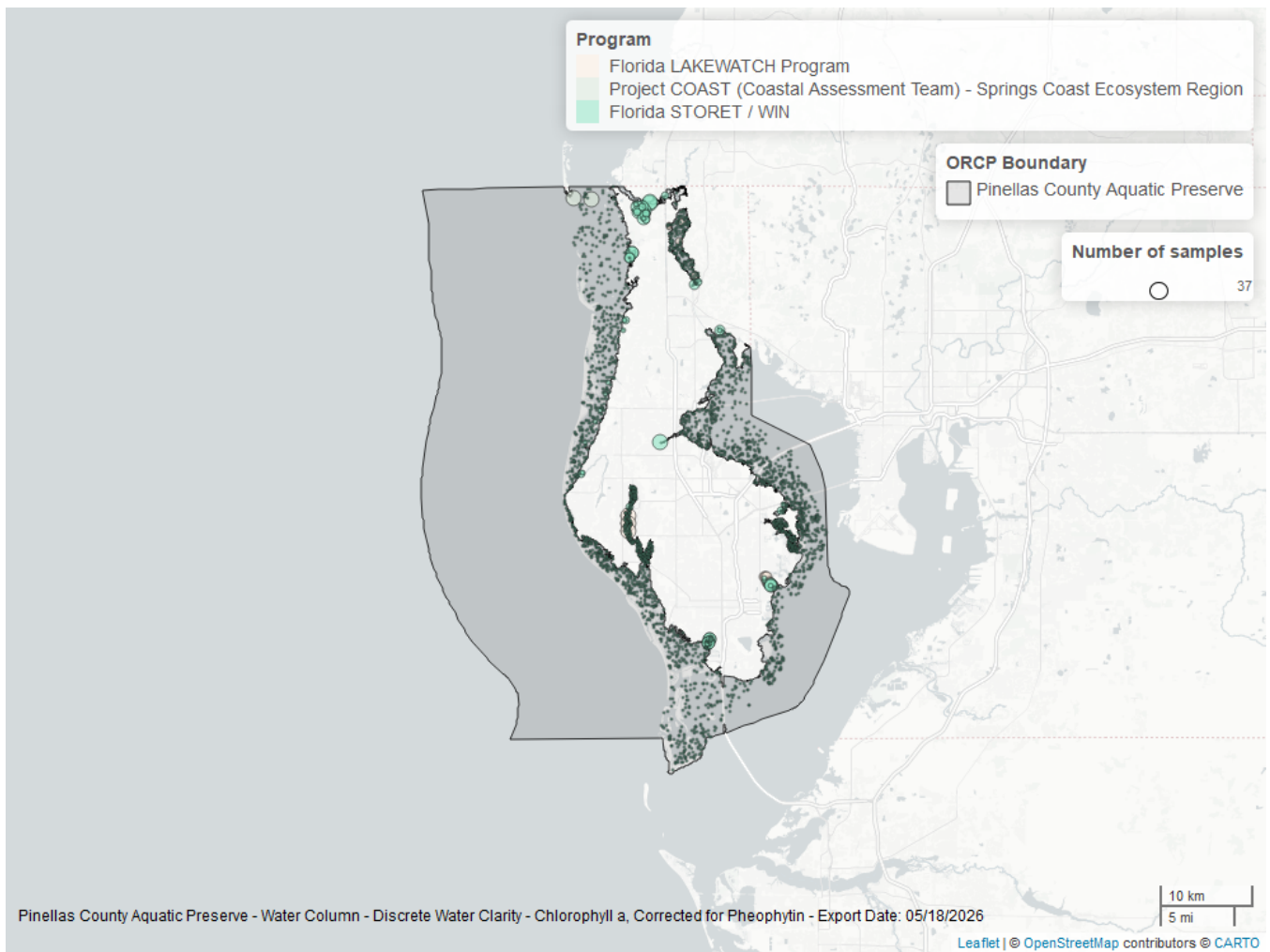


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	3372	2000	2025
514	307	2018	2025
5008	64	2023	2025

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

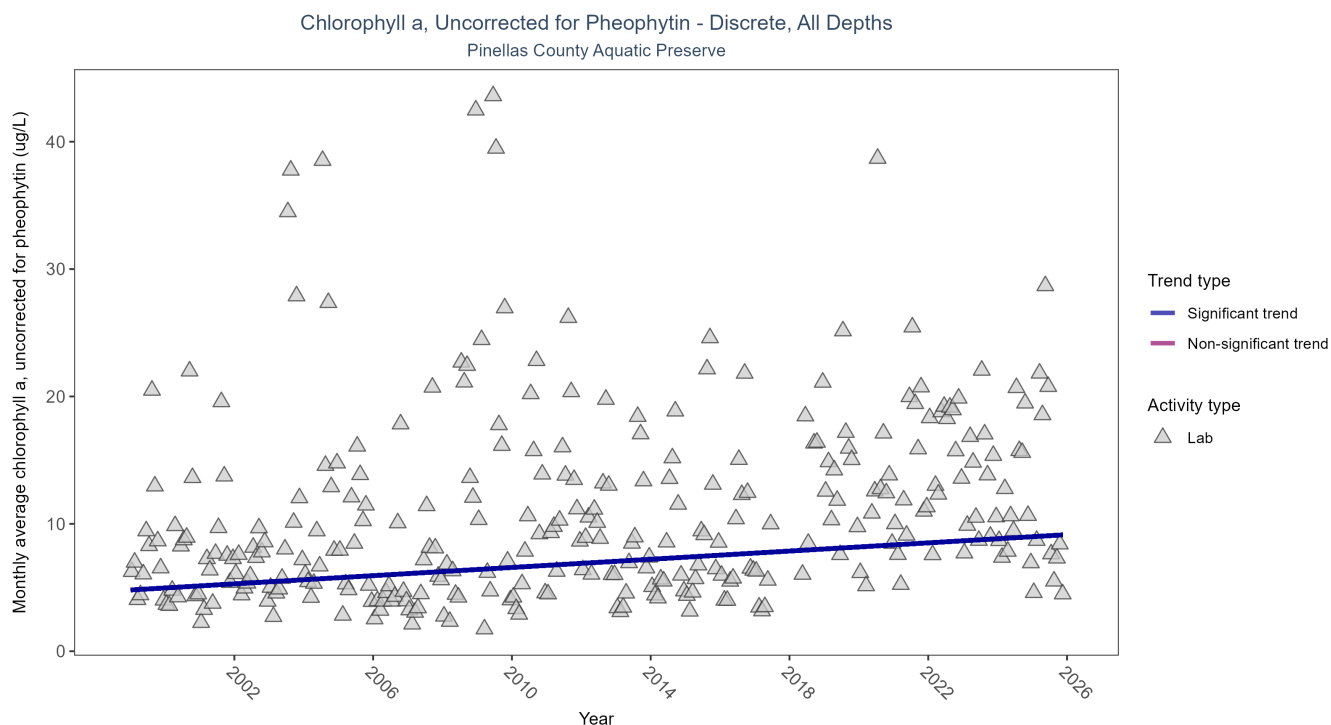


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	7148	27	1999 - 2025	6.4	0.241	4.806	0.1609	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.16 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

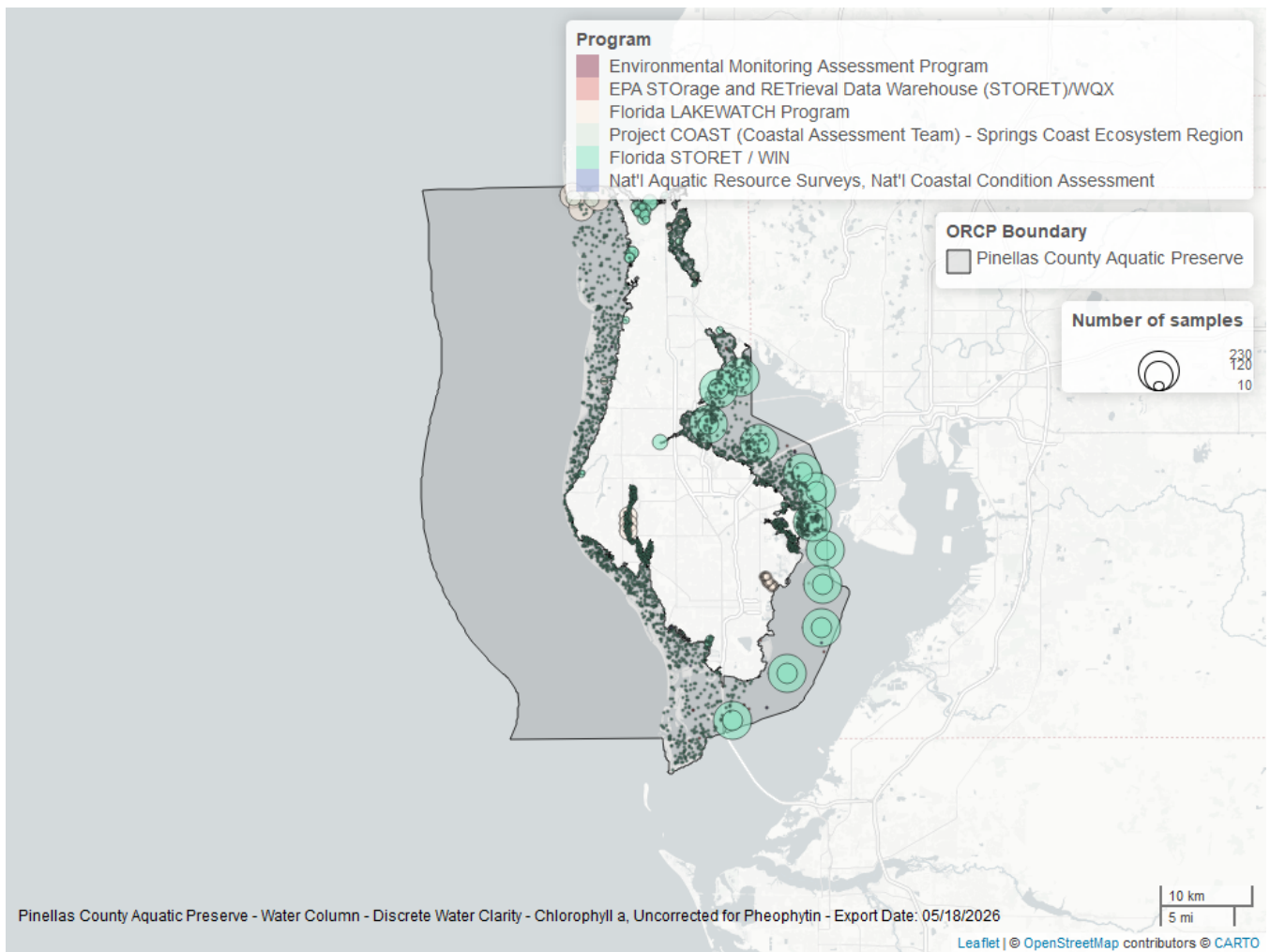


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	6275	1999	2025
514	799	2001	2025
5008	64	2023	2025
103	26	2000	2015
115	9	2000	2004
118	8	2000	2010

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

115 - Environmental Monitoring Assessment Program⁵

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁶

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

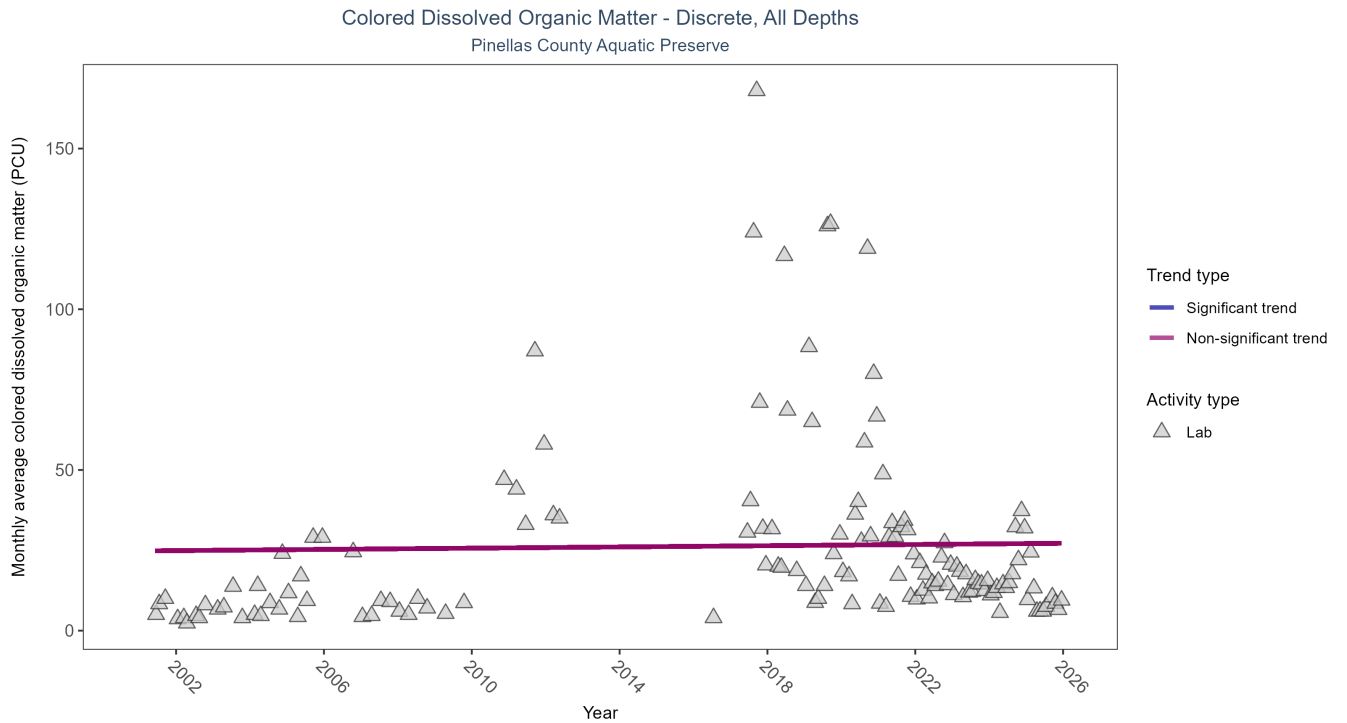


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1388	22	2001 - 2025	10	0.0065	24.8044	0.0952	0.2904

Colored dissolved organic matter showed no detectable trend between 2001 and 2025.

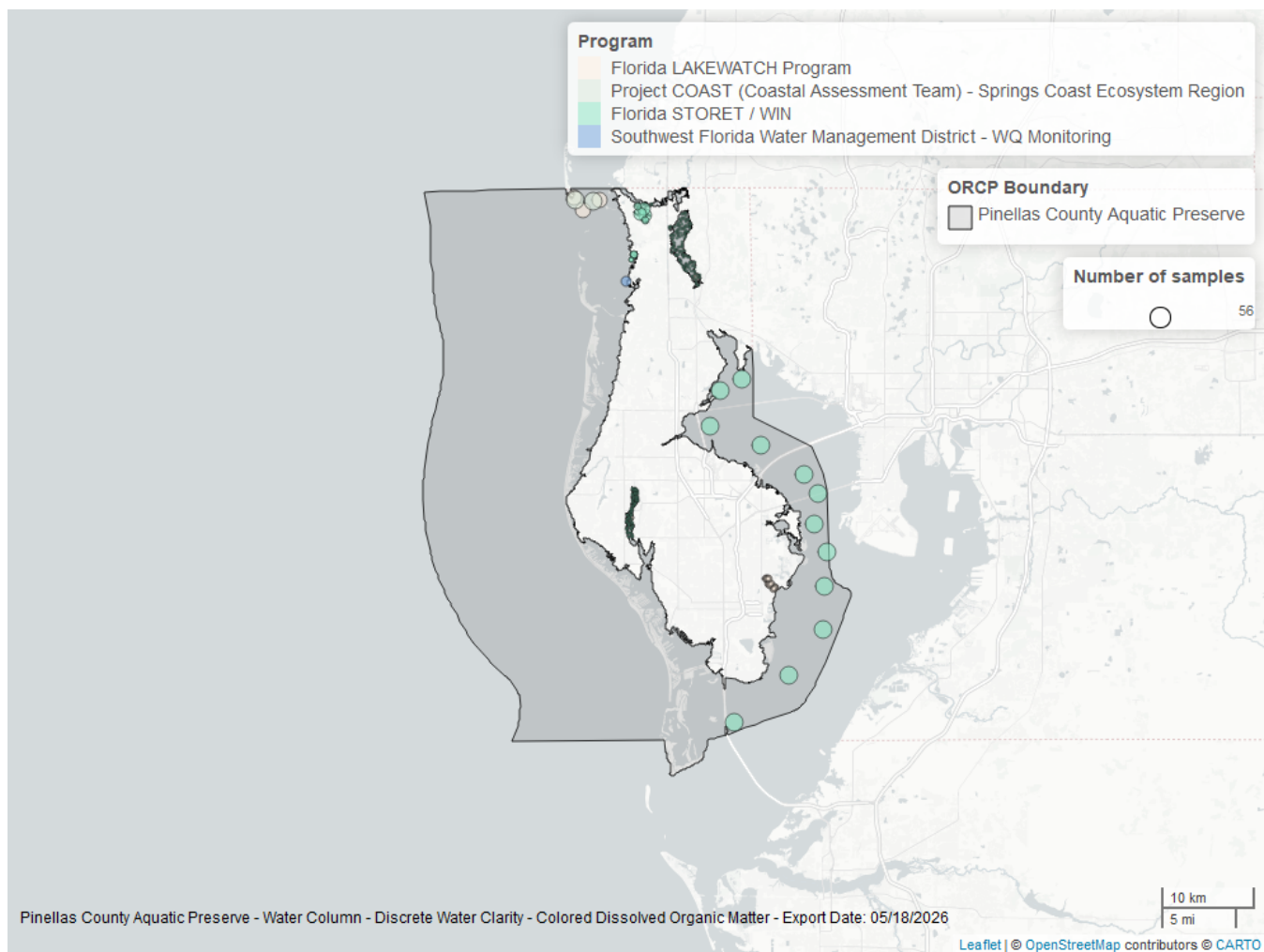


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
5002	1010	2017	2025
514	262	2001	2024
5008	112	2021	2025
479	13	2016	2024

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

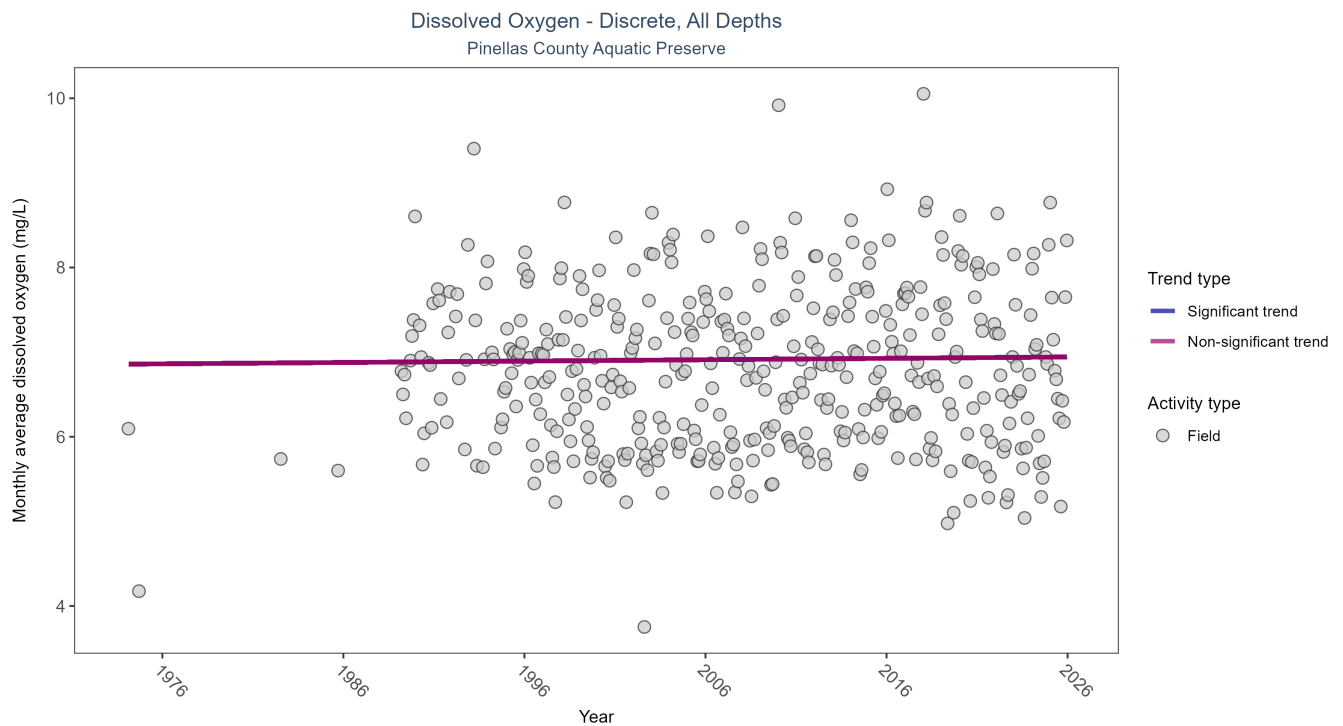


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	97306	40	1974 - 2025	6.6	0.0297	6.8582	0.0016	0.4634

Dissolved oxygen showed no detectable trend between 1974 and 2025.

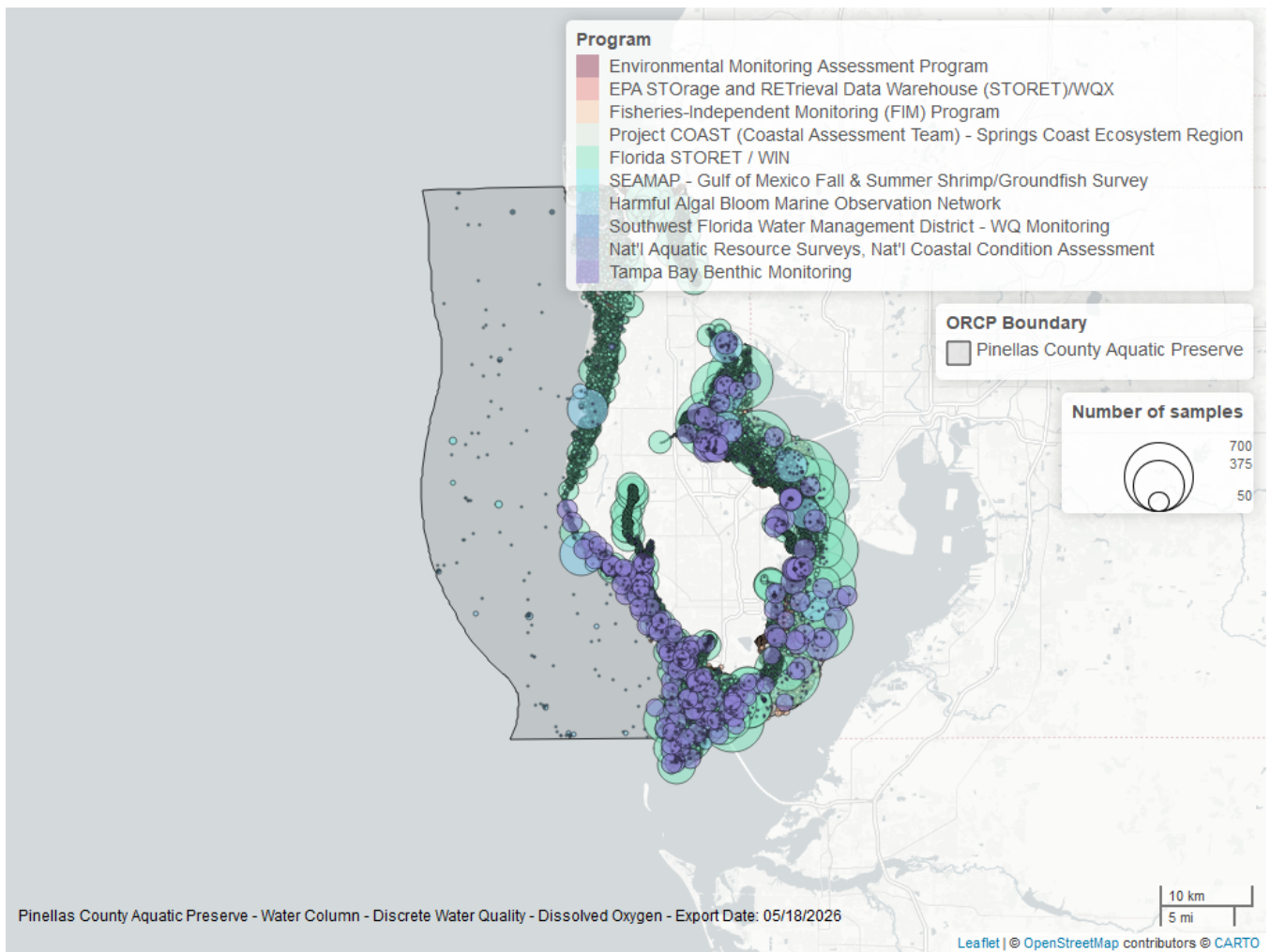


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	52597	1995	2025
69	31264	1989	2024
4067	11610	1993	2023
95	2367	1974	2018
5008	112	2021	2025
118	32	2000	2020
115	29	2000	2004
60	23	2008	2013
479	10	2016	2024
103	10	2015	2015

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁸

4067 - Tampa Bay Benthic Monitoring⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰
5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁶
115 - Environmental Monitoring Assessment Program⁵
60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey¹¹
479 - Southwest Florida Water Management District - Water Quality Monitoring⁷
103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

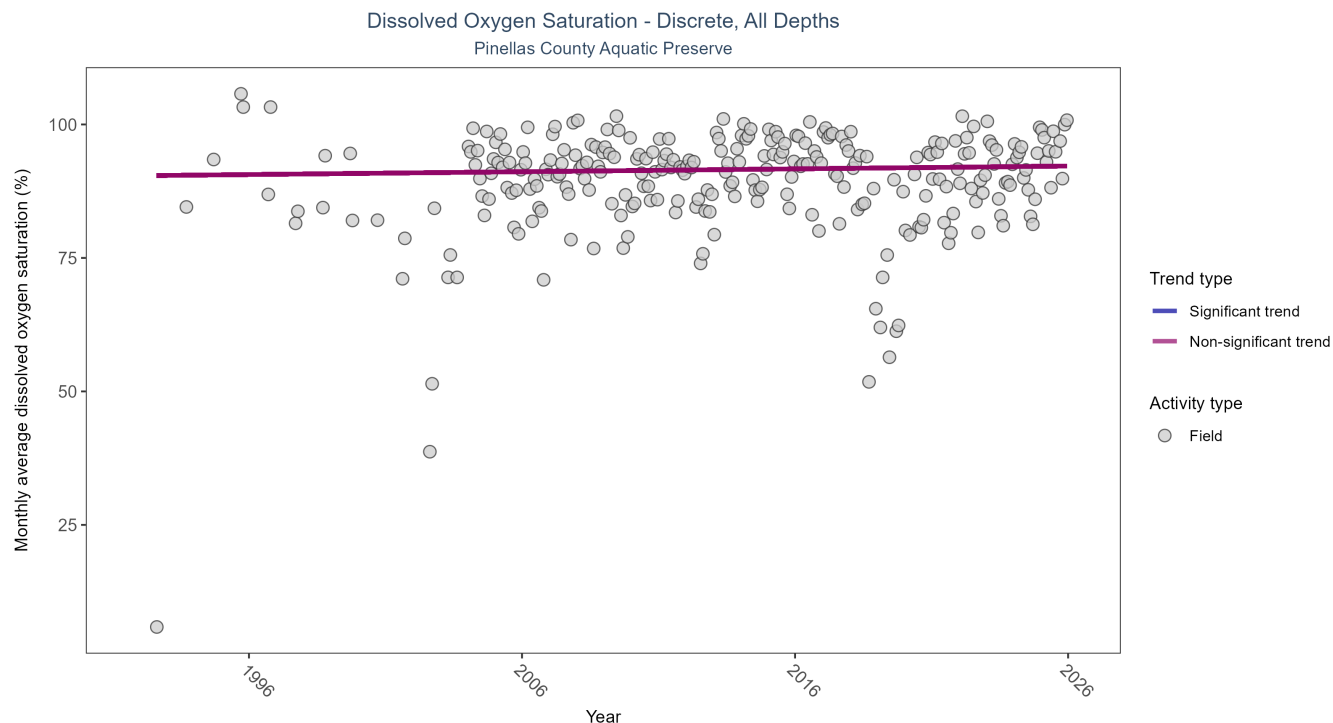


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	31862	34	1992 - 2025	91	0.0502	90.4014	0.0532	0.319

Dissolved oxygen saturation showed no detectable trend between 1992 and 2025.

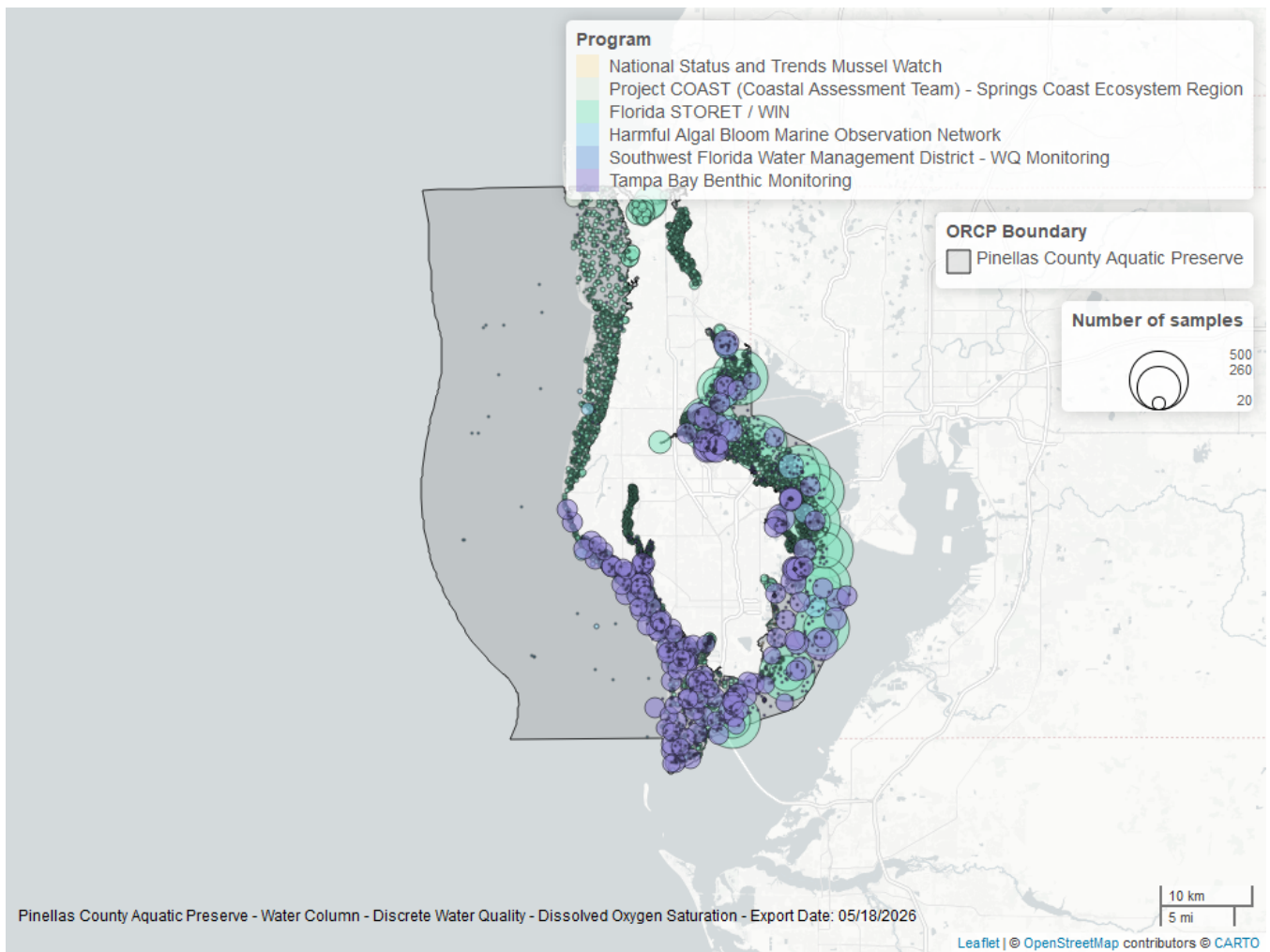


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	20546	2004	2025
4067	11046	1993	2023
95	638	2002	2018
5008	104	2021	2025
102	66	1992	1992
479	1	2024	2024

Program names:

5002 - Florida STORET / WIN¹

4067 - Tampa Bay Benthic Monitoring⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

102 - National Status and Trends Mussel Watch¹²

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

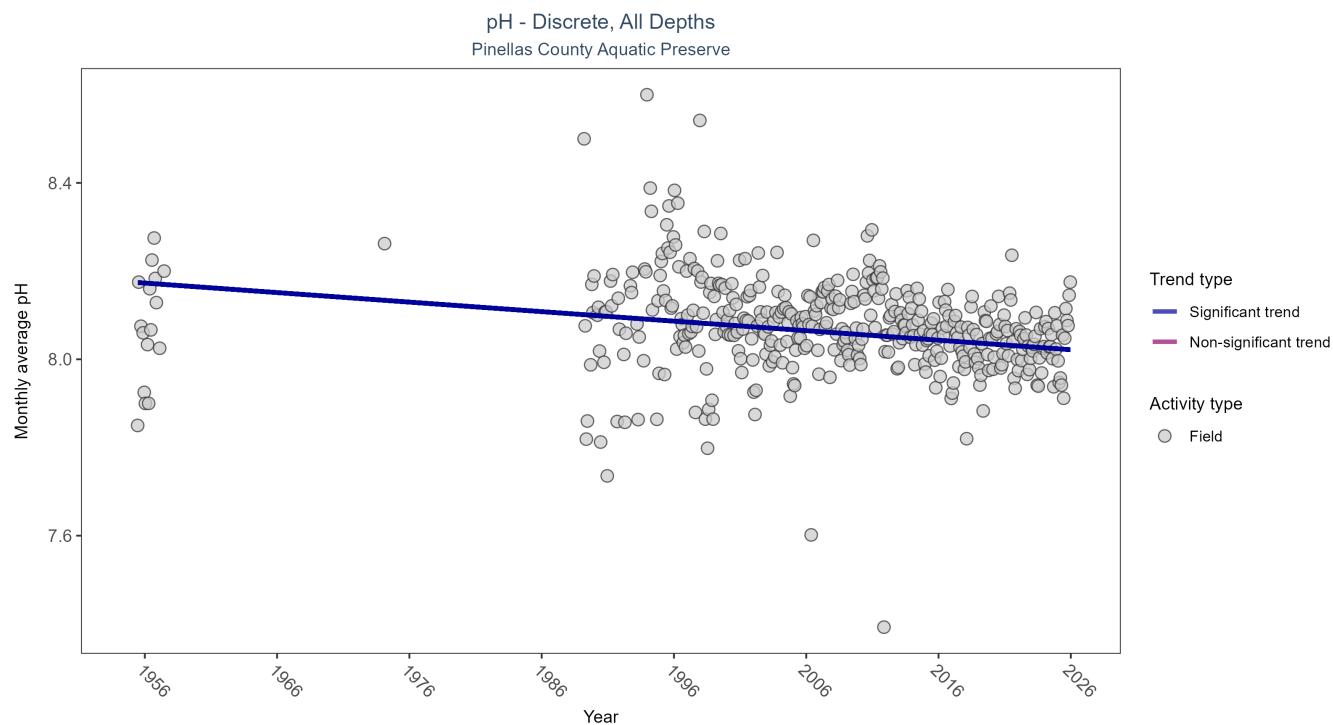


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	92528	41	1955 - 2025	8.1	-0.2069	8.175	-0.0022	0

Monthly average pH decreased by less than 0.01 pH units per year.

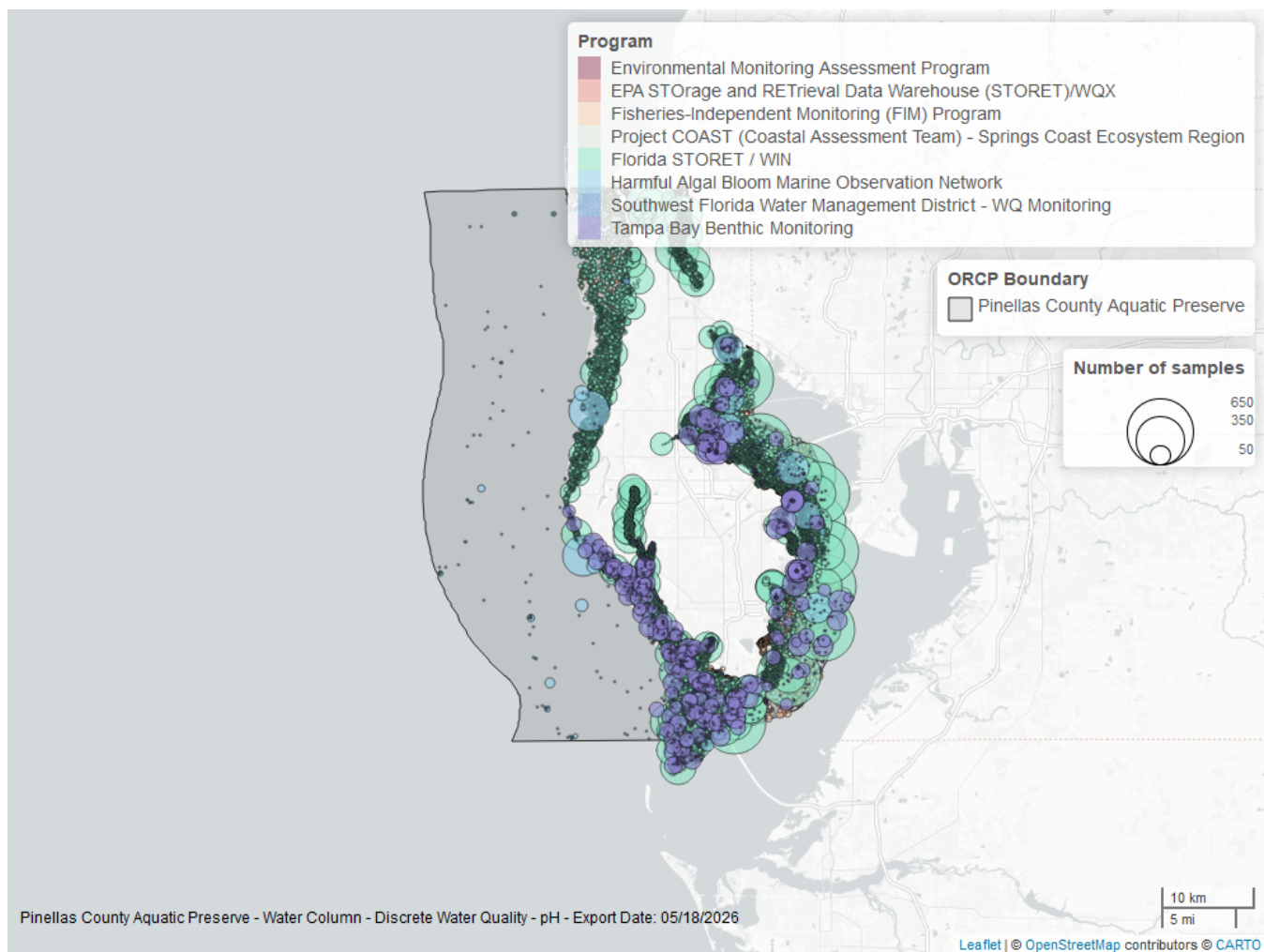


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	50556	1995	2025
69	30884	1989	2024
4067	9050	1993	2023
95	2301	1955	2018
5008	106	2021	2025
115	29	2000	2004
479	10	2016	2024
103	6	2015	2015

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁸

4067 - Tampa Bay Benthic Monitoring⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

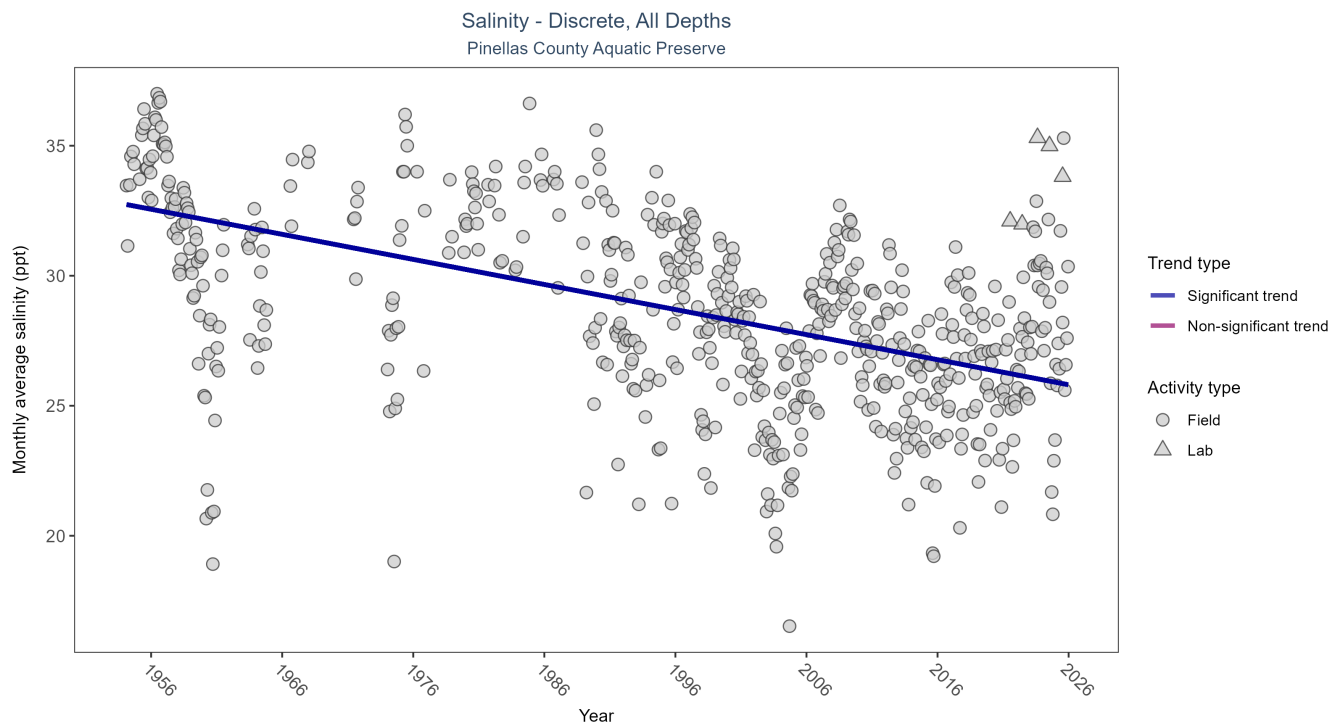


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	96816	65	1954 - 2025	28.95	-0.3608	32.7501	-0.0964	0

Monthly average salinity decreased by 0.1 ppt per year.

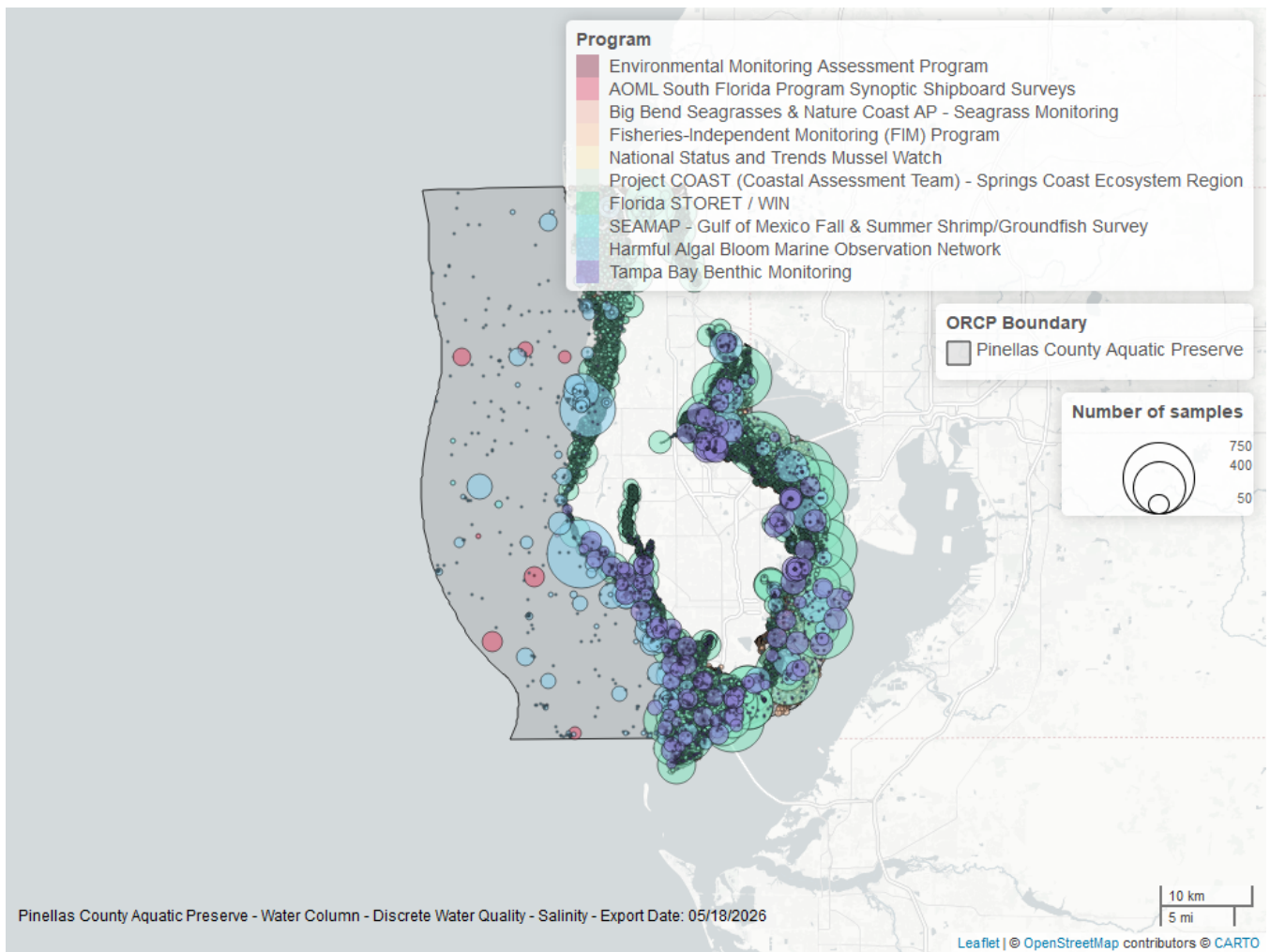


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	50443	1995	2025
69	31553	1989	2024
4067	7290	1993	2023
95	7192	1954	2018
3	288	2018	2025
5008	110	2021	2025
102	66	1992	1992
560	40	2021	2025
115	29	2000	2004
60	16	2008	2013

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁸

4067 - Tampa Bay Benthic Monitoring⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰

3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹³

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

102 - National Status and Trends Mussel Watch¹²

560 - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring¹⁴

115 - Environmental Monitoring Assessment Program⁵

60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey¹¹

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

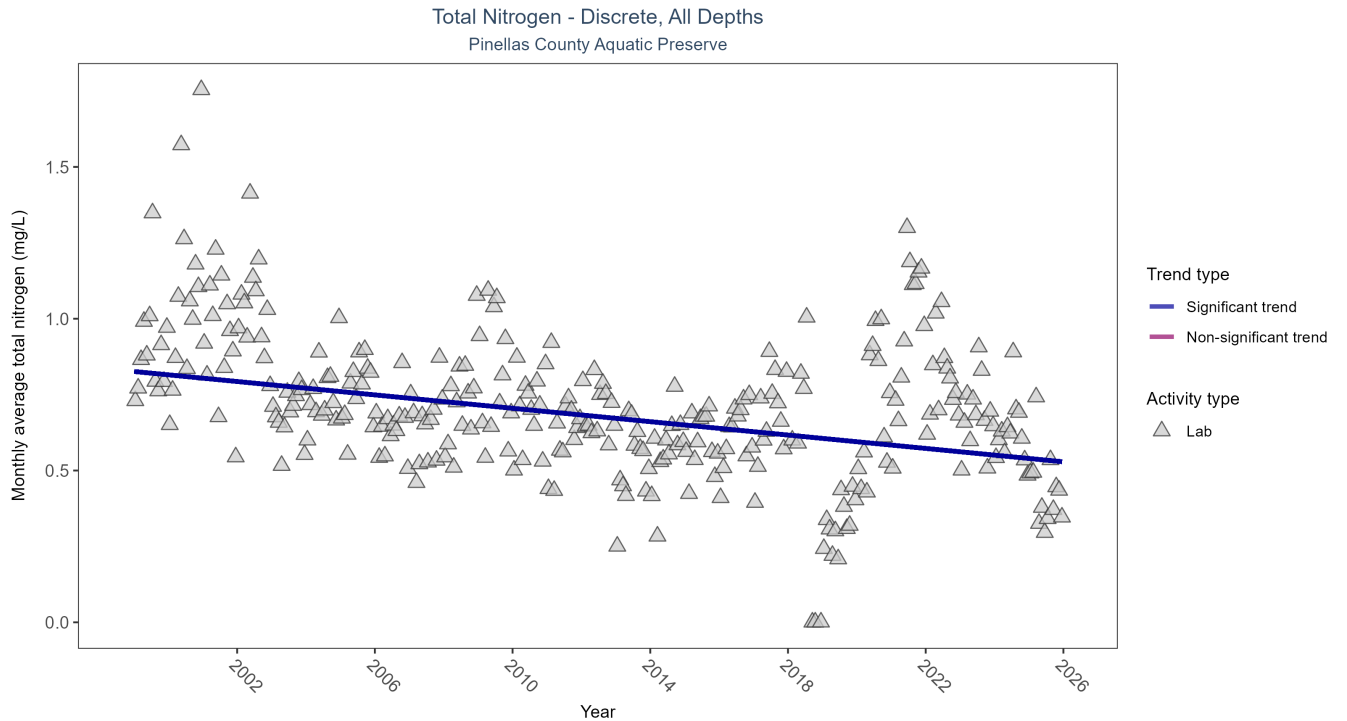


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	16755	27	1999 - 2025	0.589	-0.2907	0.8262	-0.011	0

Monthly average total nitrogen decreased by 0.01 mg/L per year.

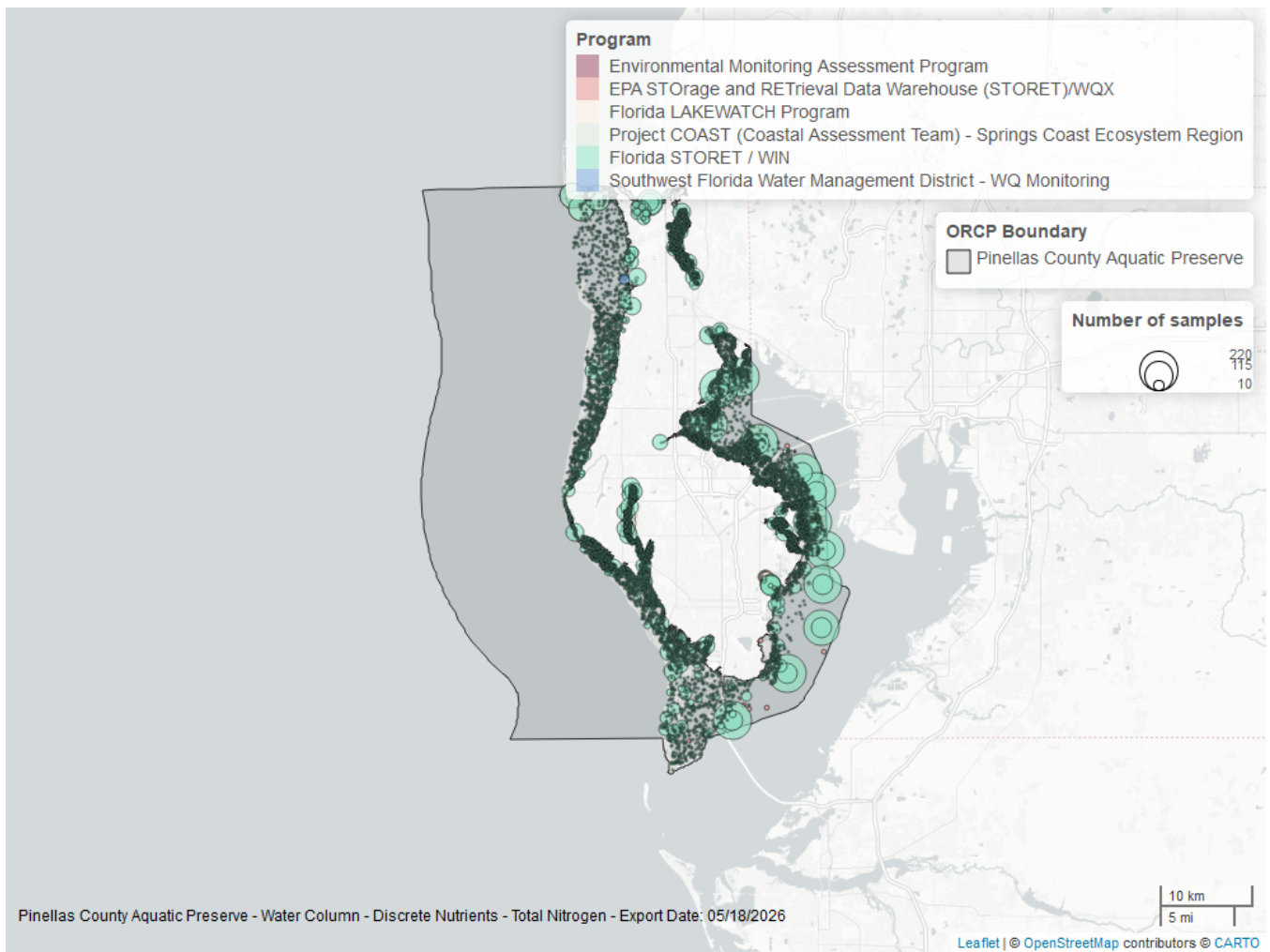


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	15806	1999	2025
514	871	2001	2024
5008	66	2023	2025
103	61	2000	2015
479	13	2016	2024
115	9	2000	2004

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

115 - Environmental Monitoring Assessment Program⁵

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

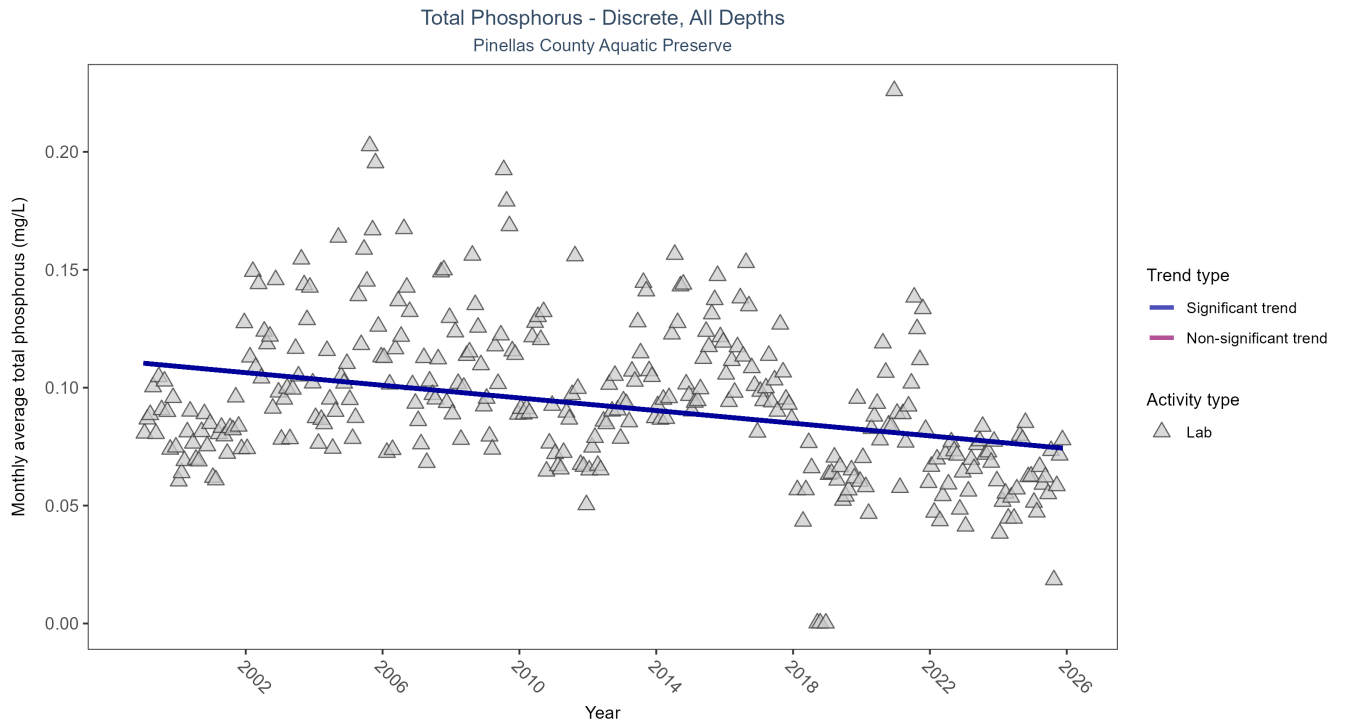


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	16551	27	1999 - 2025	0.087	-0.273	0.1104	-0.0013	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

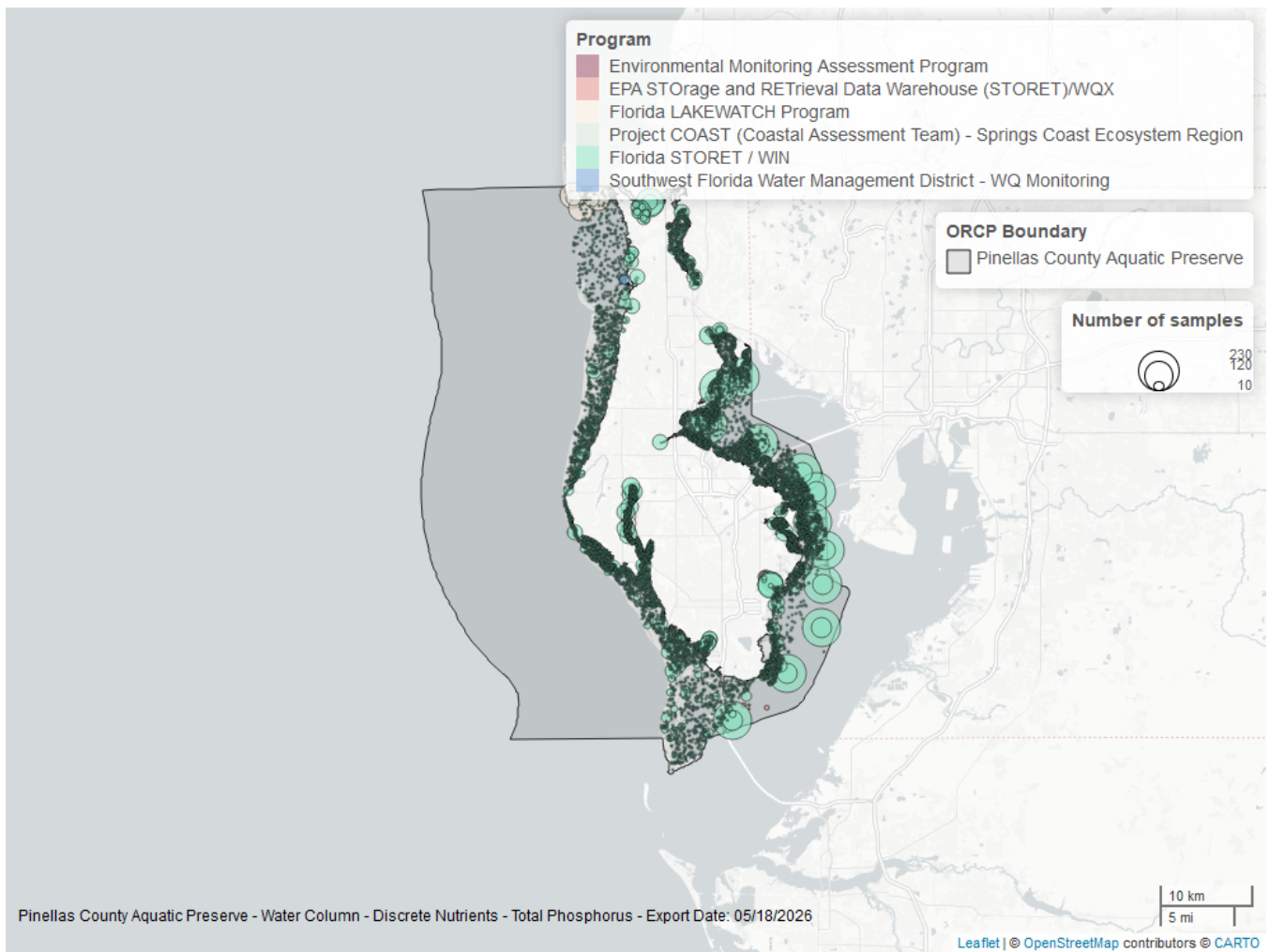


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	15716	1999	2025
514	876	2001	2024
5008	64	2023	2025
103	47	2000	2015
479	13	2016	2024
115	9	2000	2004

Program names:

5002 - Florida STORET / WIN¹

514 - Florida LAKEWATCH Program²

5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

115 - Environmental Monitoring Assessment Program⁵

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

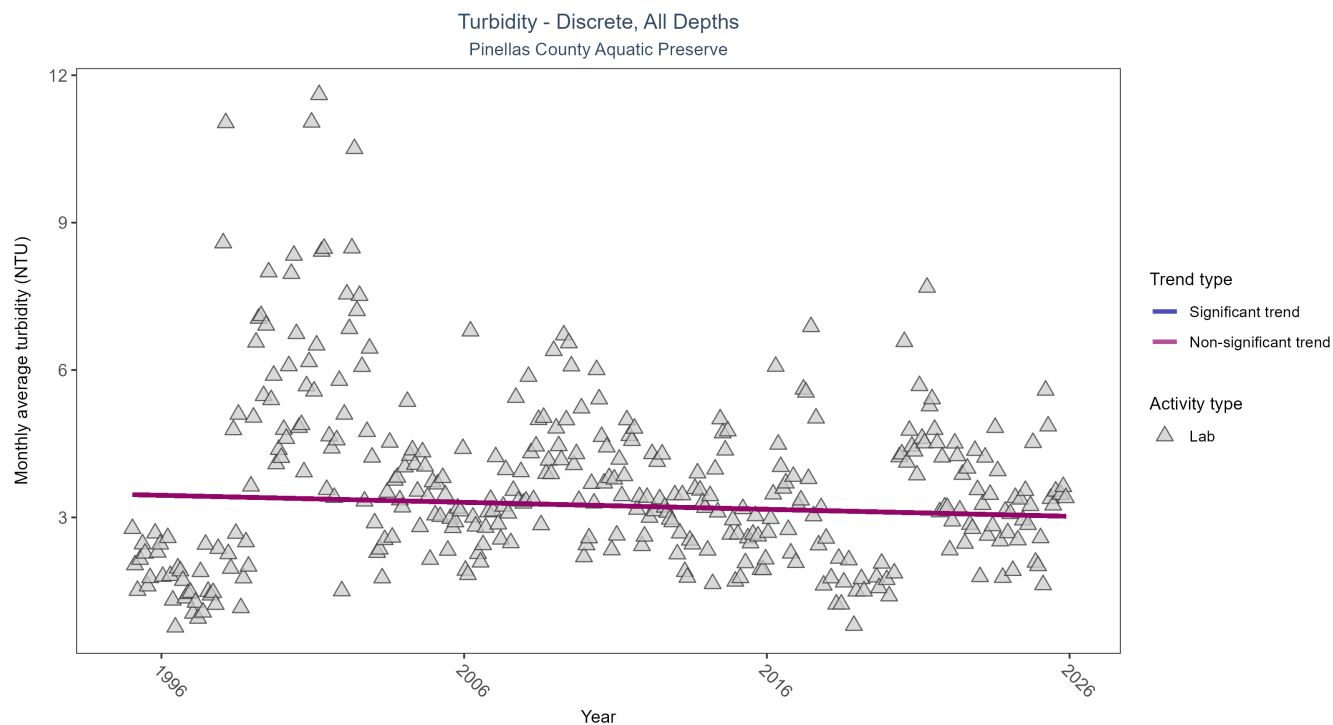


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	21801	31	1995 - 2025	2.5	-0.0519	3.4637	-0.0142	0.1751

Turbidity showed no detectable trend between 1995 and 2025.

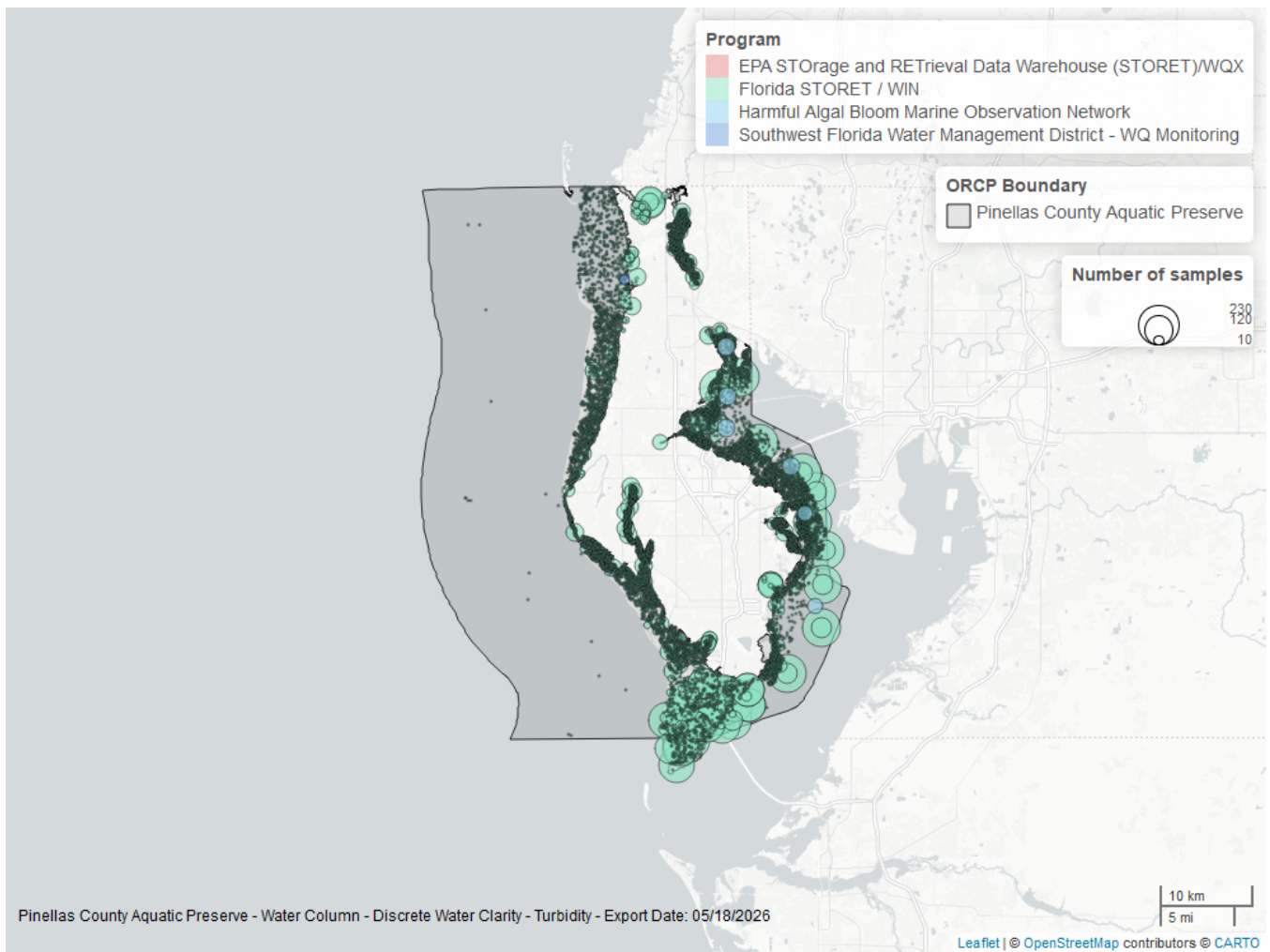


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	22591	1995	2025
95	323	2000	2016
479	13	2016	2024
103	4	2006	2006

Program names:

5002 - Florida STORET / WIN¹

95 - Harmful Algal Bloom Marine Observation Network¹⁰

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

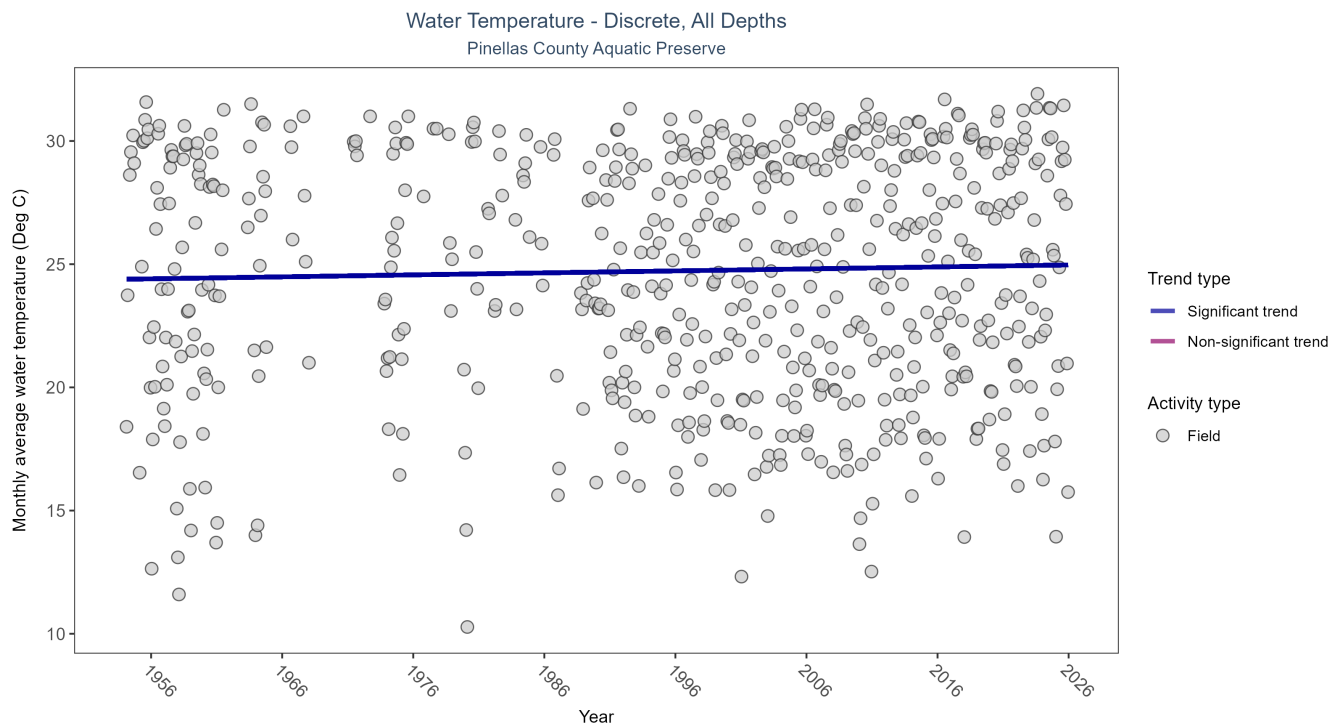


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	103247	68	1954 - 2025	26.55	0.0898	24.3892	0.008	0.0016

Monthly average water temperature increased by 0.01°C per year.

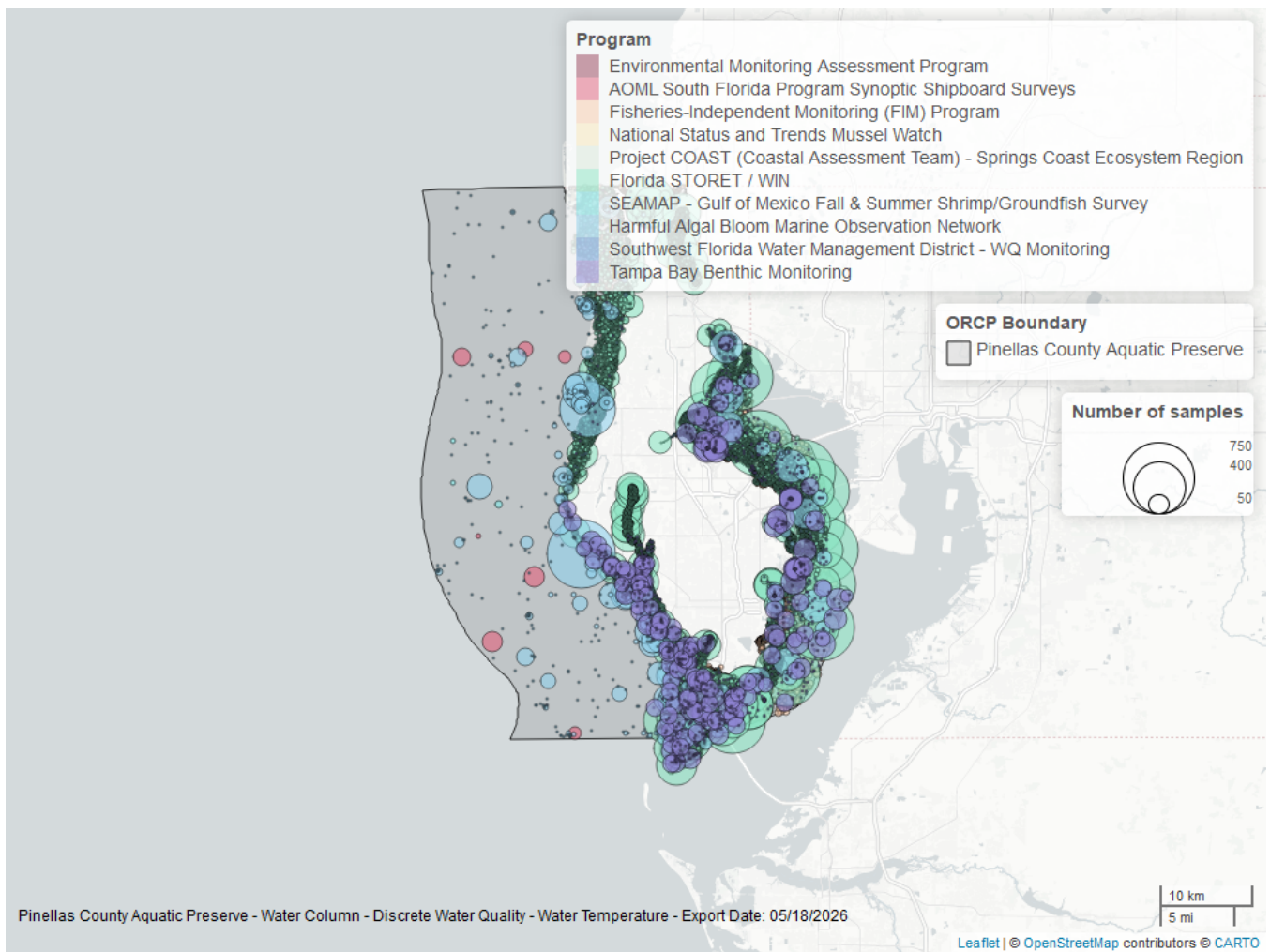


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	53062	1995	2025
69	31612	1989	2024
4067	10489	1993	2023
95	7592	1954	2018
3	280	2018	2025
5008	106	2021	2025
102	66	1992	1992
115	29	2000	2004
60	19	2008	2013
479	10	2016	2024

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁸

4067 - Tampa Bay Benthic Monitoring⁹

95 - Harmful Algal Bloom Marine Observation Network¹⁰
3 - Atlantic Oceanographic and Meteorological Laboratory (AOML) South Florida Program Synoptic Shipboard Surveys¹³
5008 - Project COAST (Coastal Assessment Team) - Springs Coast Ecosystem Region³
102 - National Status and Trends Mussel Watch¹²
115 - Environmental Monitoring Assessment Program⁵
60 - Southeast Area Monitoring and Assessment Program (SEAMAP) - Gulf of Mexico Fall & Summer Shrimp/Groundfish Survey¹¹
479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Pinellas County Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
5	CWBF1	21	TRUE	Water Temperature
5	FHPF1	4	FALSE	Water Temperature
7	02310175	4	FALSE	Water Temperature , Specific Conductivity
7	02310207	3	FALSE	Water Temperature , Specific Conductivity

Program names:

5 - National Data Buoy Center¹⁵

7 - National Water Information System¹⁶

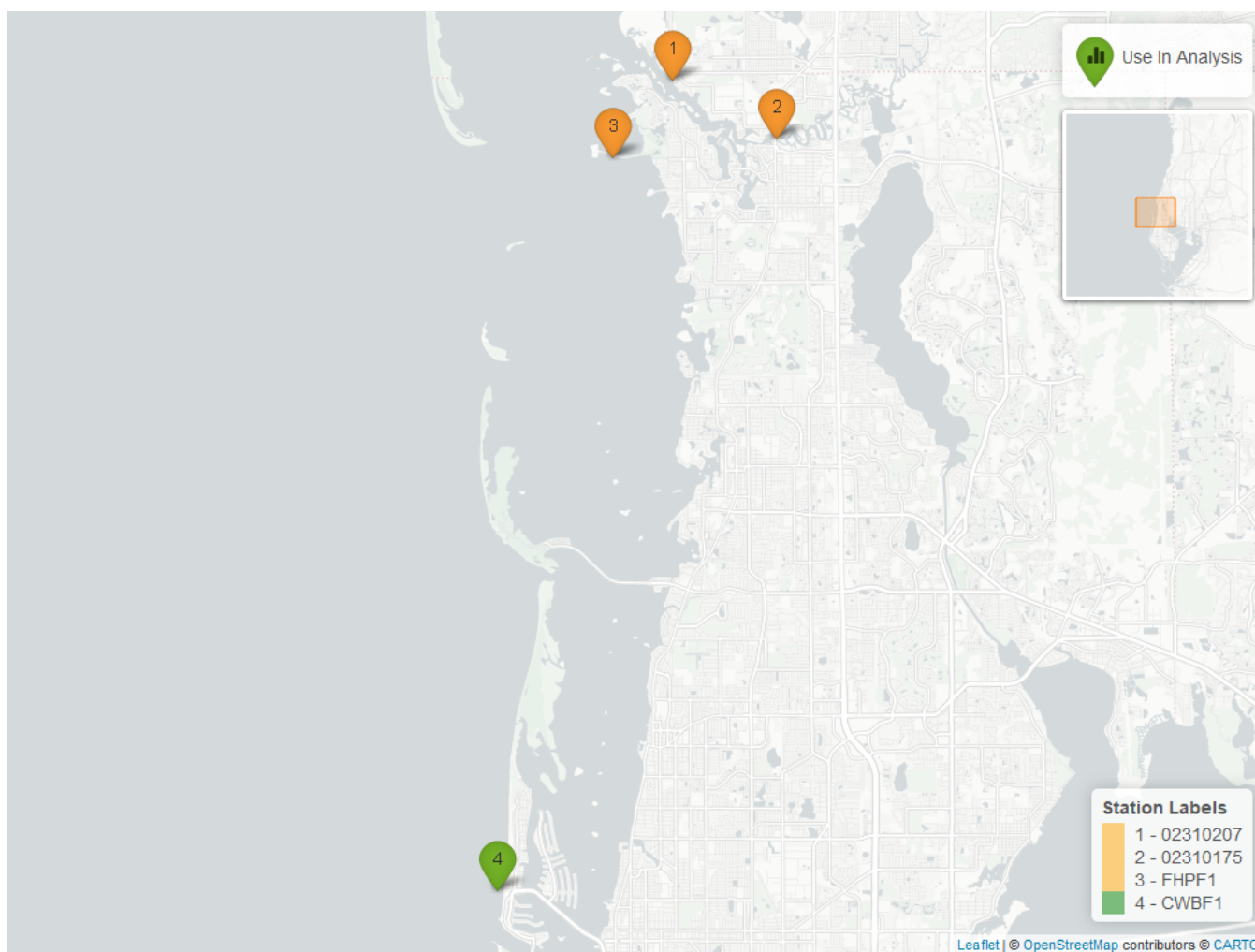


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

Water Temperature - Continuous

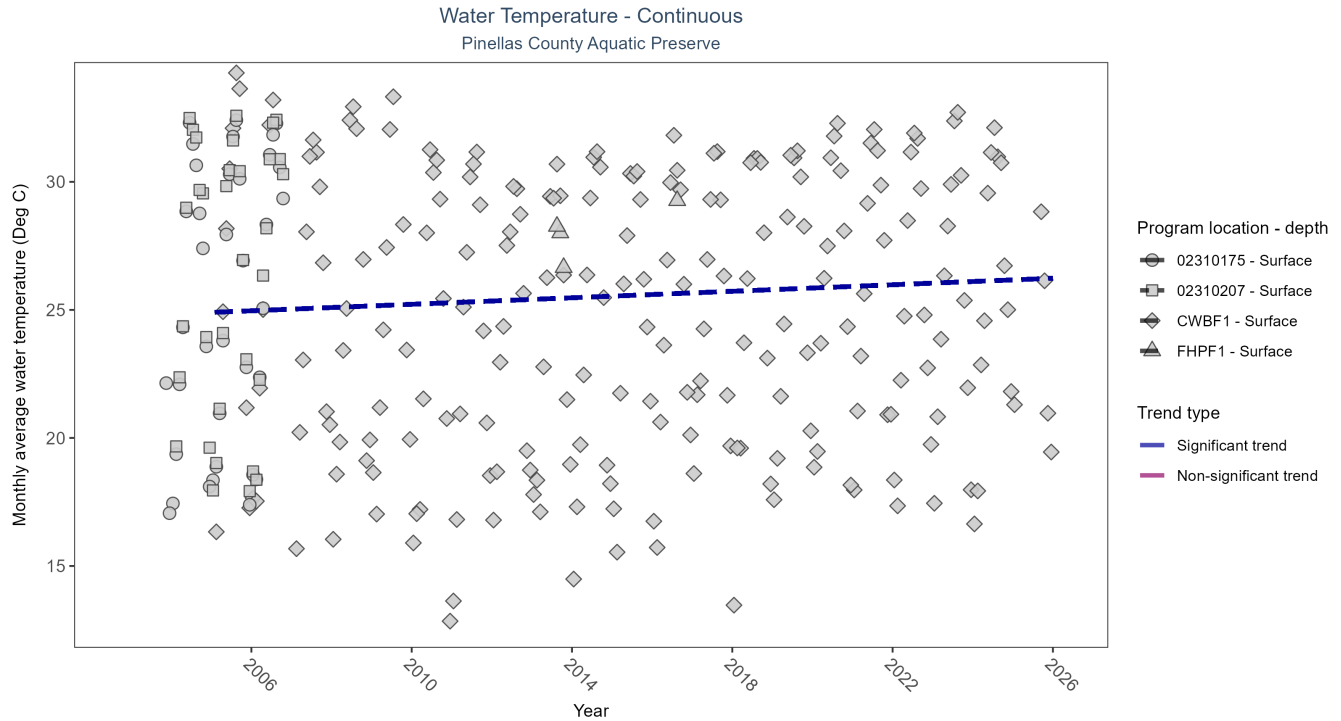


Figure 24: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CWBF1	Significantly increasing trend	1443248	21	2005 - 2025	25.40	0.18	24.9	0.06	0
FHPPF1	Insufficient data to calculate trend	12636	2	2013 - 2016	27.90	-	-	-	-
02310175	Insufficient data to calculate trend	1421	4	2003 - 2006	26.50	-	-	-	-
02310207	Insufficient data to calculate trend	1424	3	2004 - 2006	27.55	-	-	-	-

At one program location, monthly average water temperature increased by 0.06°C per year. There was insufficient data to fit a model for three locations.

Specific Conductivity - Continuous

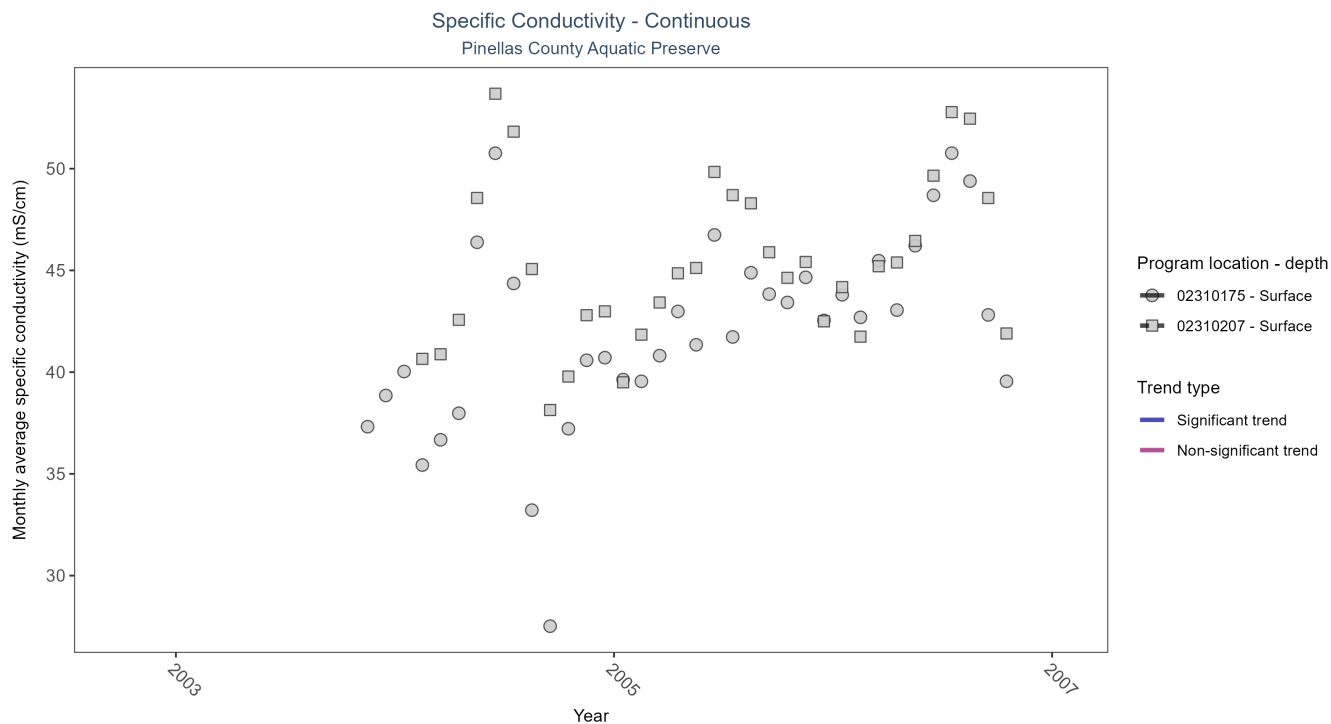


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02310175	Insufficient data to calculate trend	1876	4	2003 - 2006	42.3	-	-	-	-
02310207	Insufficient data to calculate trend	1502	3	2004 - 2006	45.6	-	-	-	-

There was insufficient data to fit a model for two locations.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Pinellas County Aquatic Preserve
SAV Percent Cover - Sample Locations

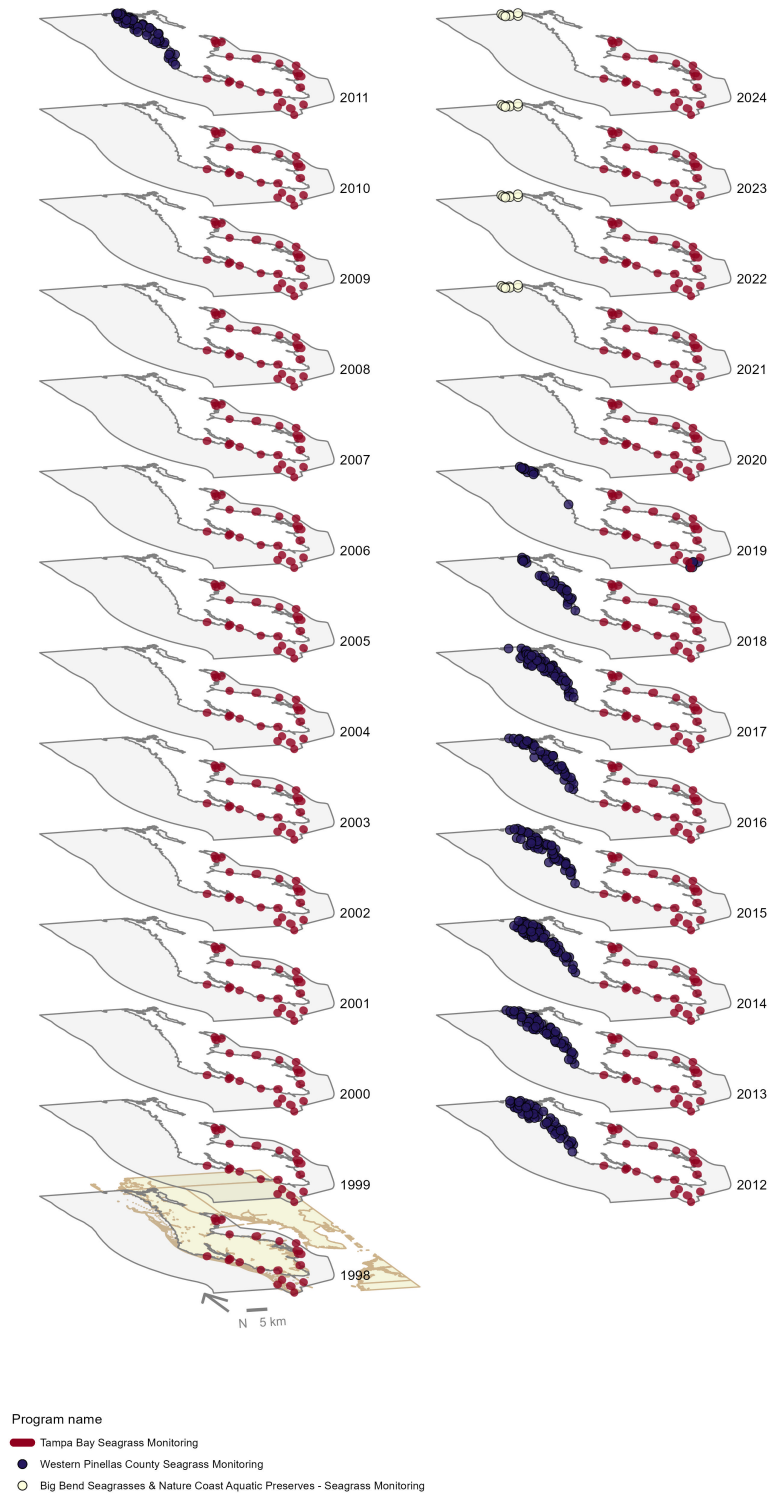


Figure 26: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Pinellas County Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

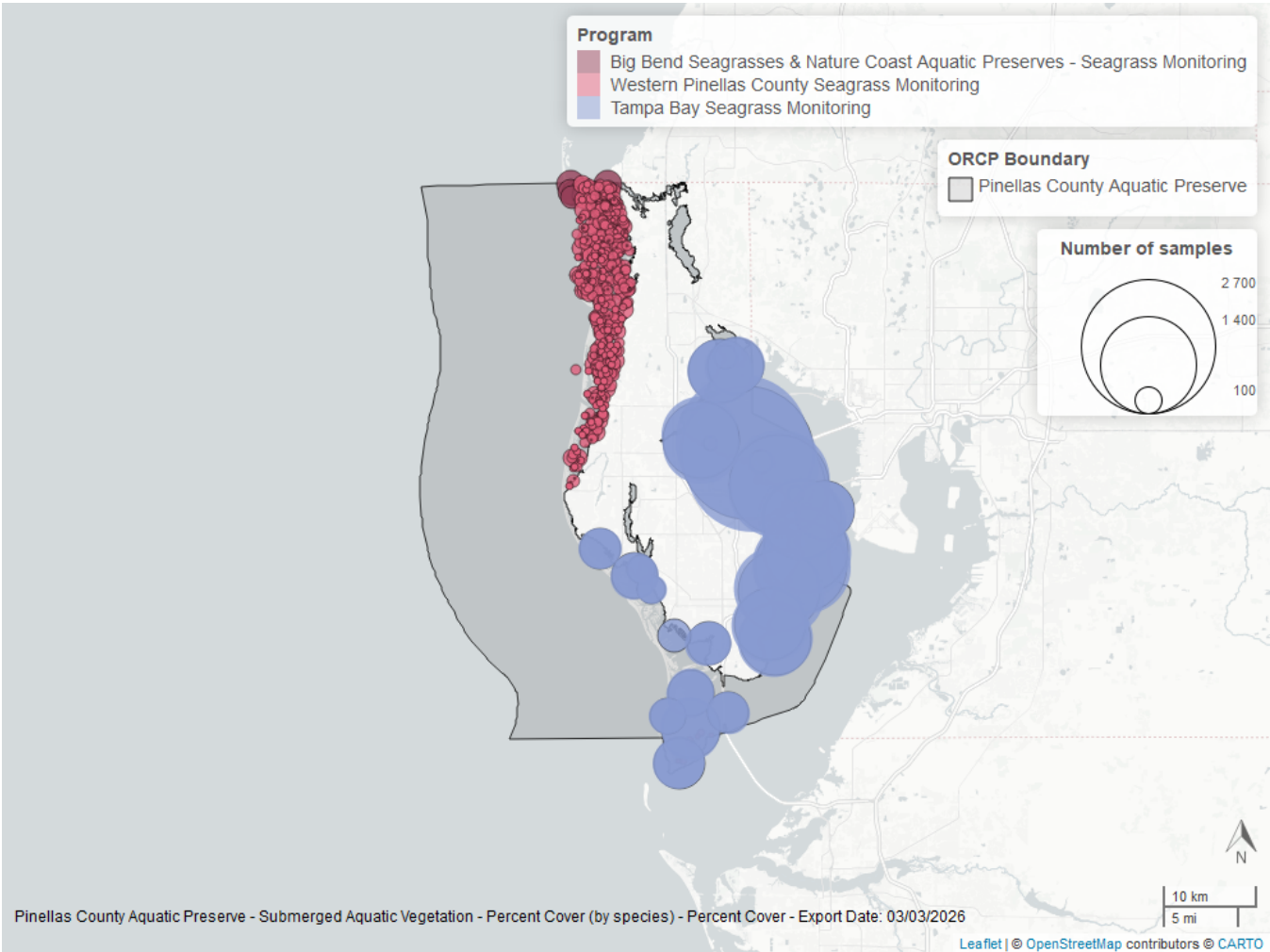


Figure 27: Map showing SAV sampling sites within the boundaries of *Pinellas County Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 31: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
565	15964	1998	2024	Braun Blanquet	27
560	435	2021	2025	Modified Braun Blanquet	8
560	416	2021	2025	Percent Cover	8
564	8342	2011	2019	Percent Cover	441

Program names:

560 - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring¹⁴

564 - Western Pinellas County Seagrass Monitoring¹⁷

565 - Tampa Bay Seagrass Monitoring¹⁸

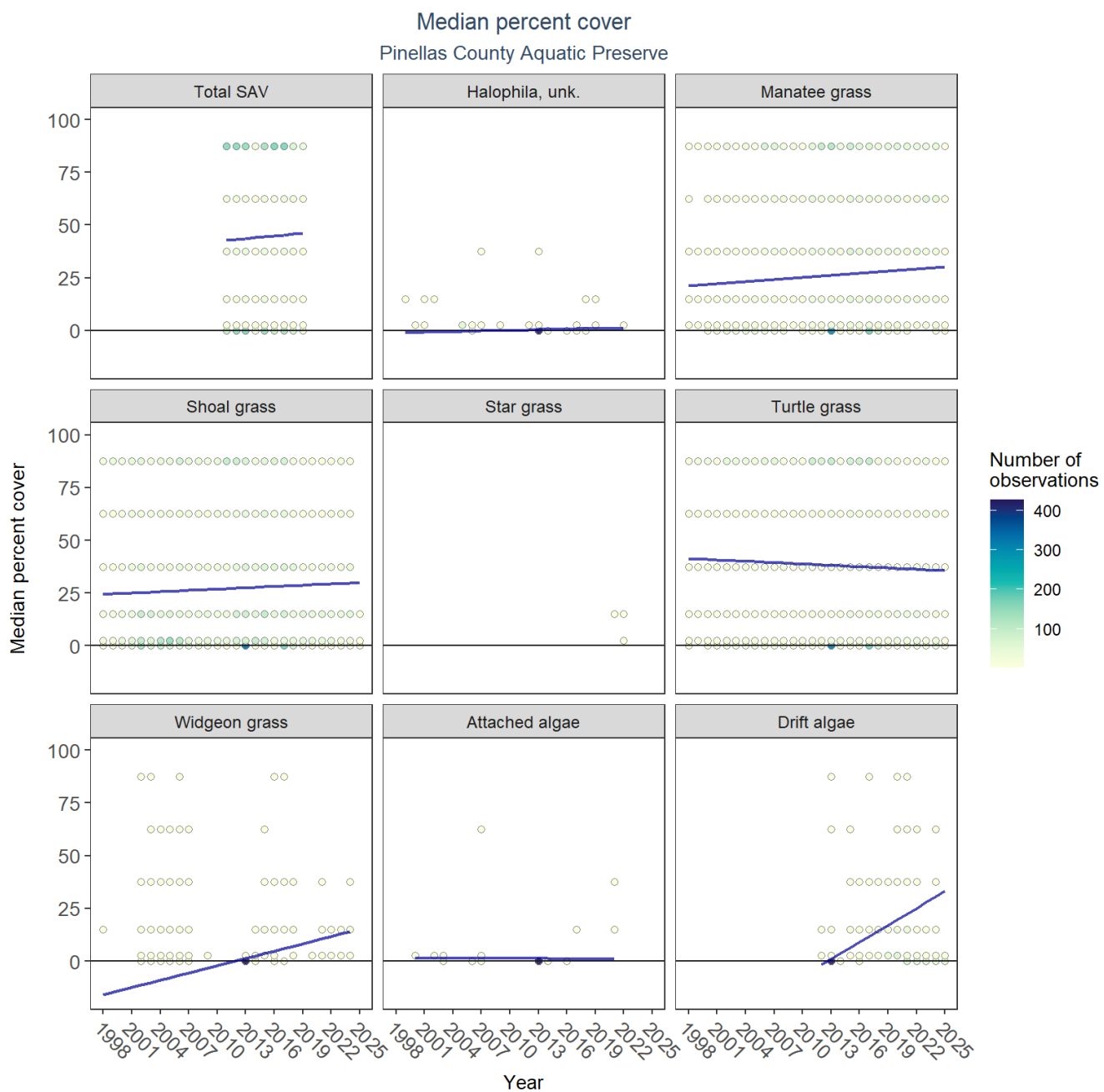


Figure 28: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

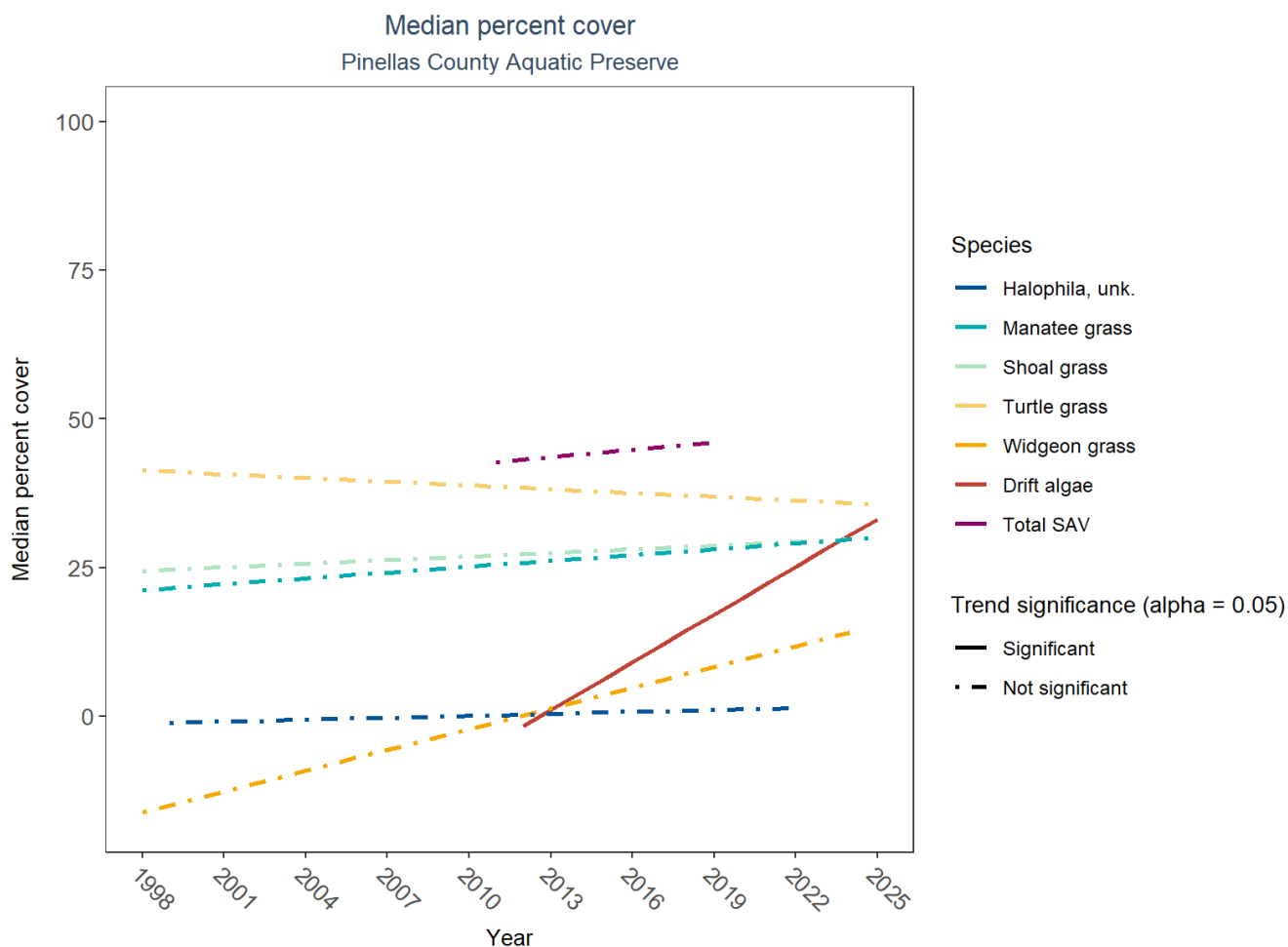


Figure 29: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 32: Percent Cover Trend Analysis for Pinellas County Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	2000 - 2021	1.346388	-0.0087228	0.9563576
Drift algae	Increasing trend	2012 - 2025	-49.761290	2.6737343	0.0000032
Shoal grass	No detectable trend	1998 - 2025	23.582598	0.2059045	0.1457132
Star grass	Insufficient data	-	-	-	-
Widgeon grass	No detectable trend	1998 - 2024	-20.717911	1.1603476	0.3683592
Manatee grass	No detectable trend	1998 - 2025	19.931486	0.3268018	0.1143153
Turtle grass	No detectable trend	1998 - 2025	42.215116	-0.2121977	0.1965870
Total SAV	No detectable trend	2011 - 2019	35.605421	0.4199155	0.5741278
Halophila, unk.	No detectable trend	1999 - 2022	-1.636642	0.1093123	0.6581673

An annual increase in percent cover was observed for drift algae (2.67%). Total SAV, unknown Halophila, manatee grass, shoal grass, turtle grass, widgeon grass, and attached algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass due to insufficient data.

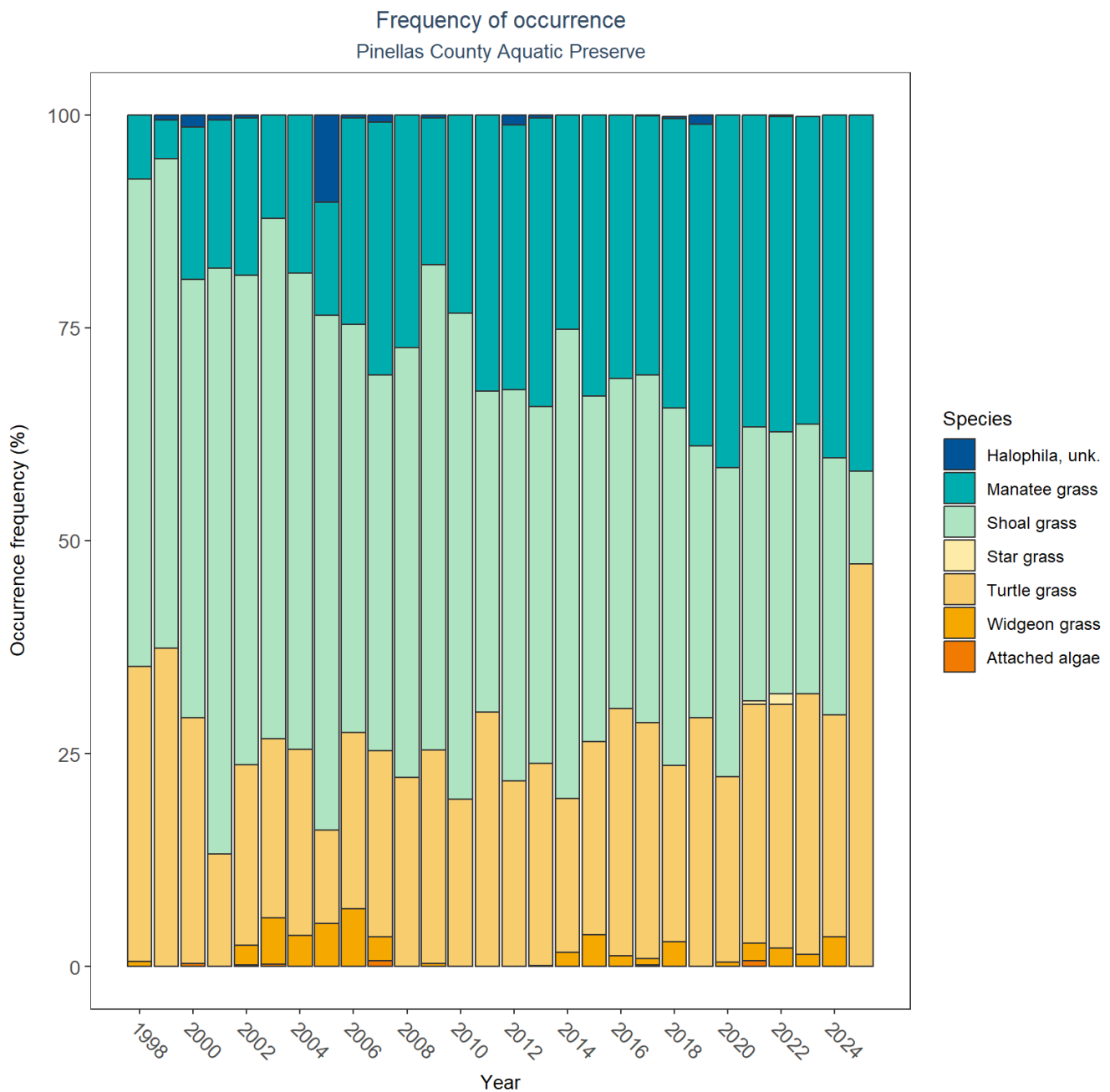


Figure 30: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Pinellas County Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

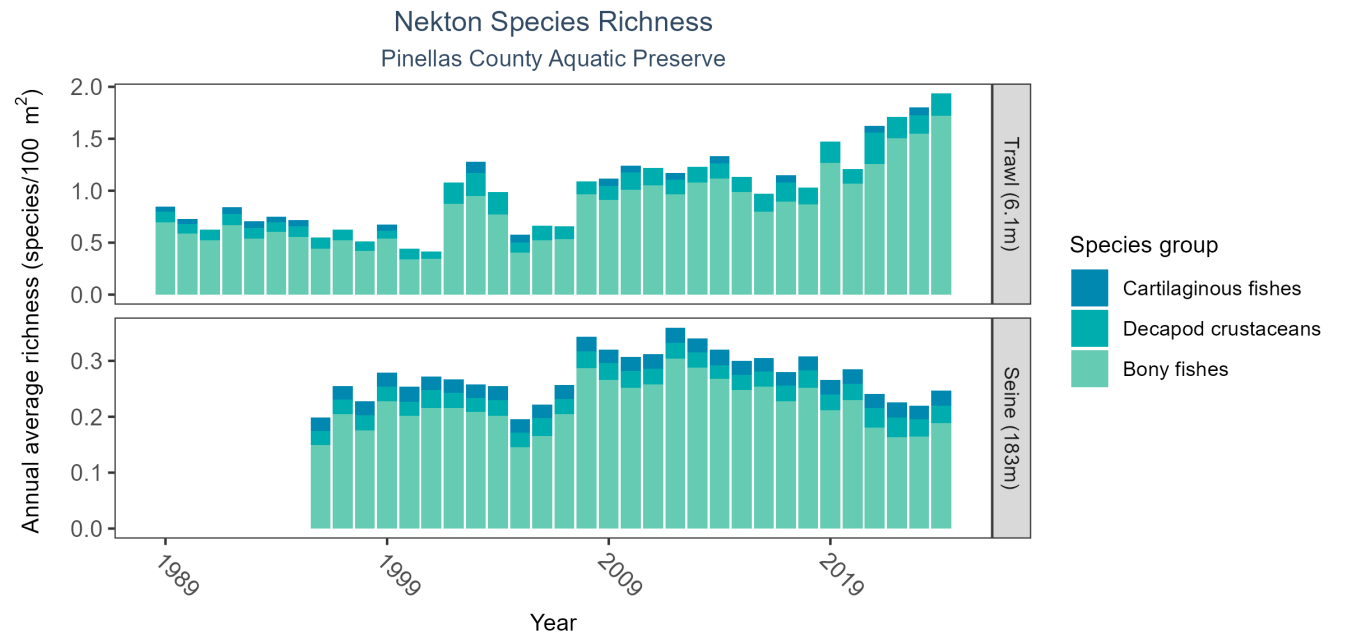


Figure 31: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 33: Nekton Species Richness

<i>Gear Type</i>	<i>No. of Samples</i>	<i>No. of Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Trawl (6.1 m)	5561	36	1989 - 2024	0.3	0.57
Seine (183 m)	4688	29	1996 - 2024	0.1	0.15

The median annual number of taxa was 0.10 based on 4,688 observations collected by 183-meter seine between 1996 and 2024, and the median annual number of taxa was 0.30 based on 5,561 observations collected by 6.1-meter trawl between 1989 and 2024.

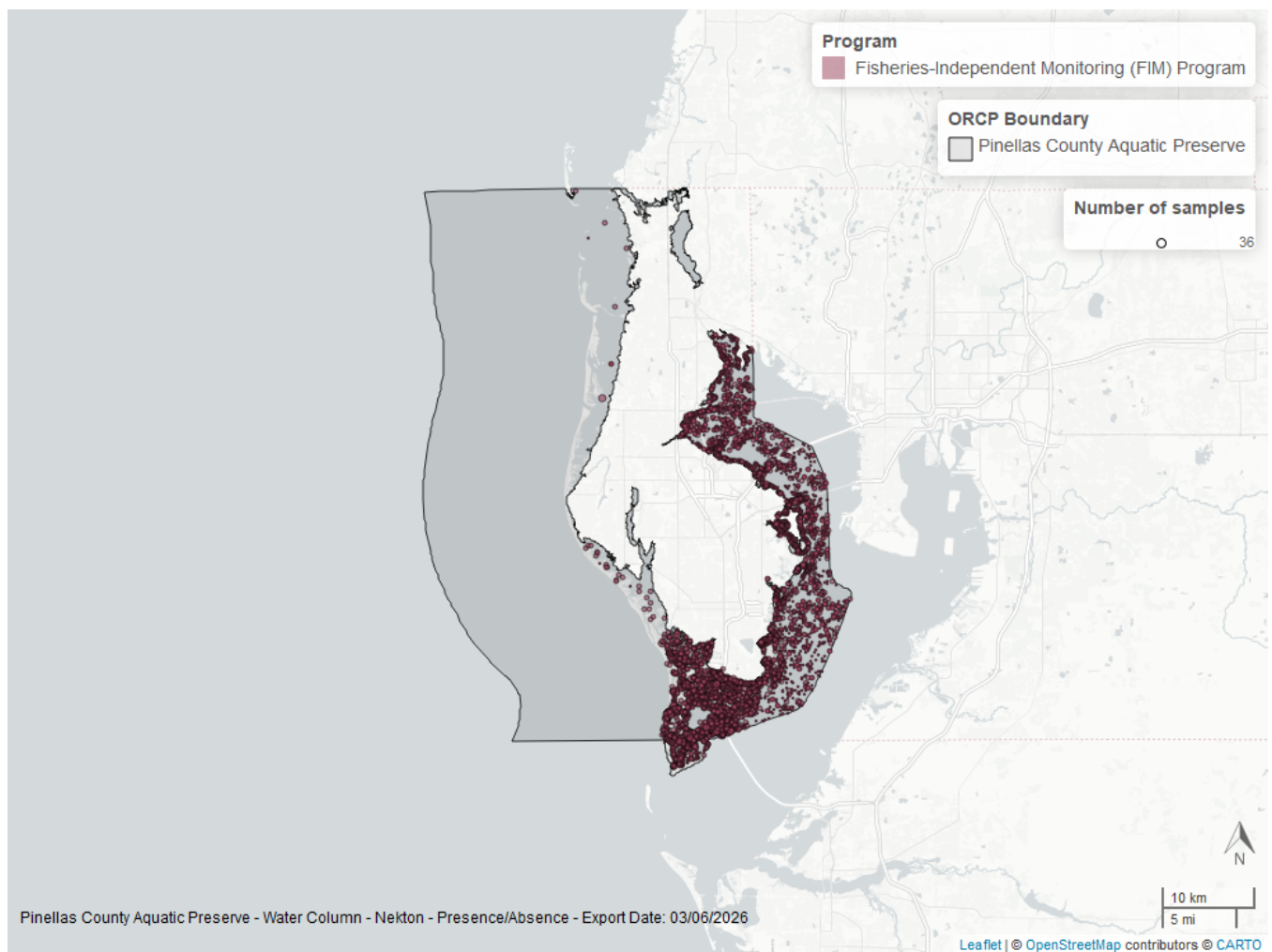


Figure 32: Map showing location of nekton sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Coastal Wetlands

The data file used is: All_CW_Parameters-2026-Jun-04.txt

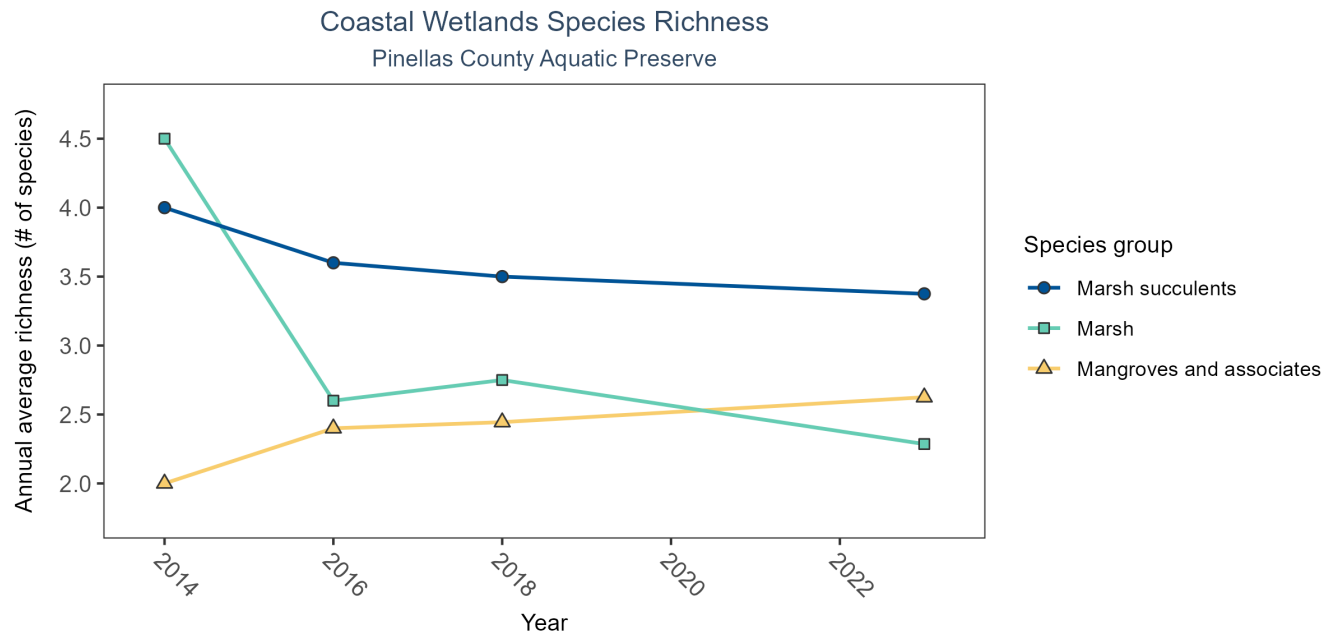


Figure 33: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 34: Coastal Wetlands Species Richness

Species Group	No. of Samples	No. Years with Data	Period of Record	Median No. of Taxa	Mean No. of Taxa
Mangroves and associates	24	4	2014 - 2023	2	2.46
Marsh	22	4	2014 - 2023	2	2.73
Marsh succulents	23	4	2014 - 2023	4	3.52

Between 2014 and 2023, the median annual number of species for mangroves and associates was 2 based on 24 observations. Between 2014 and 2023, the median annual number of species for marsh was 2 based on 22 observations. Between 2014 and 2023, the median annual number of species for marsh succulents was 4 based on 23 observations.

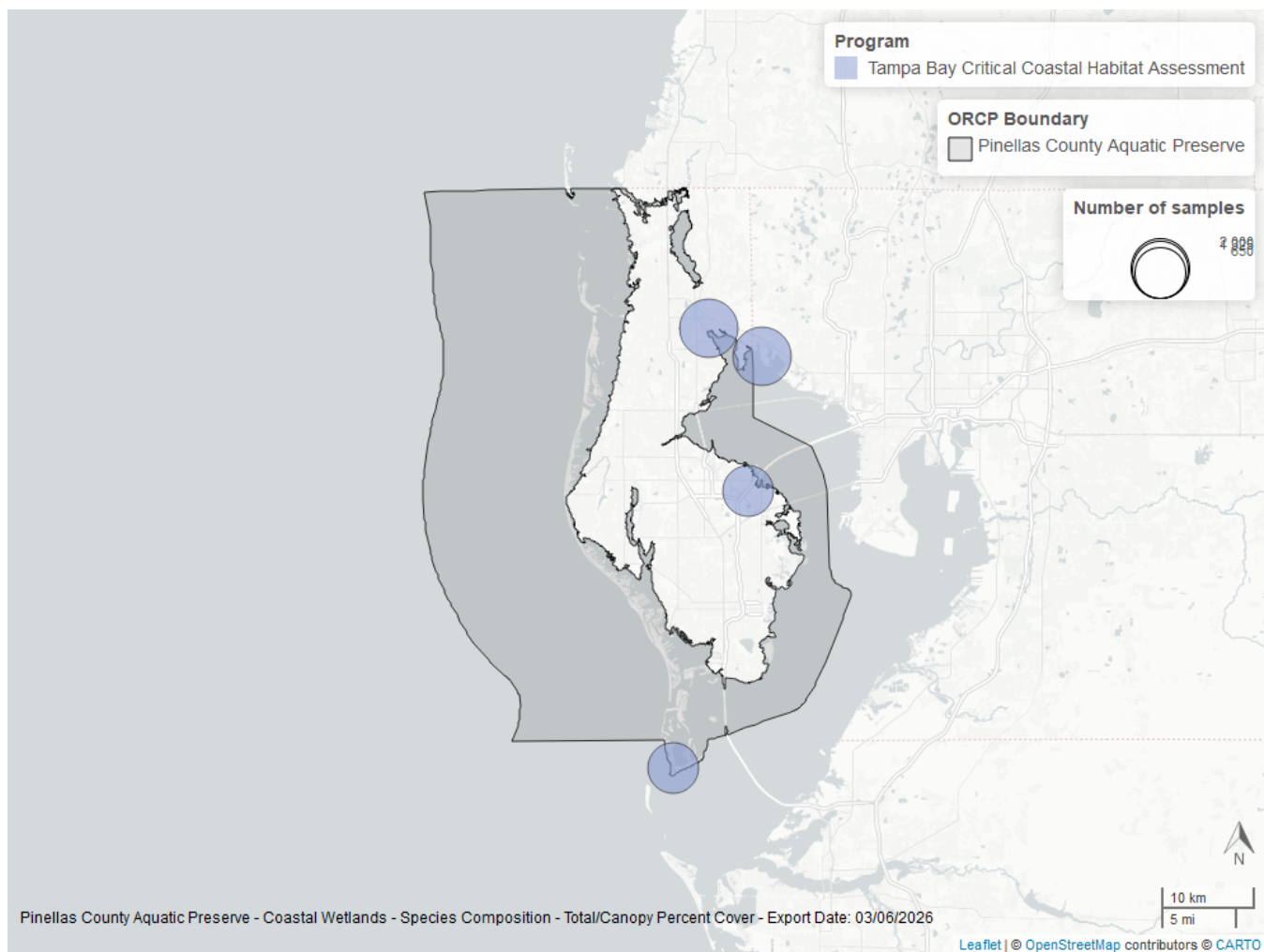


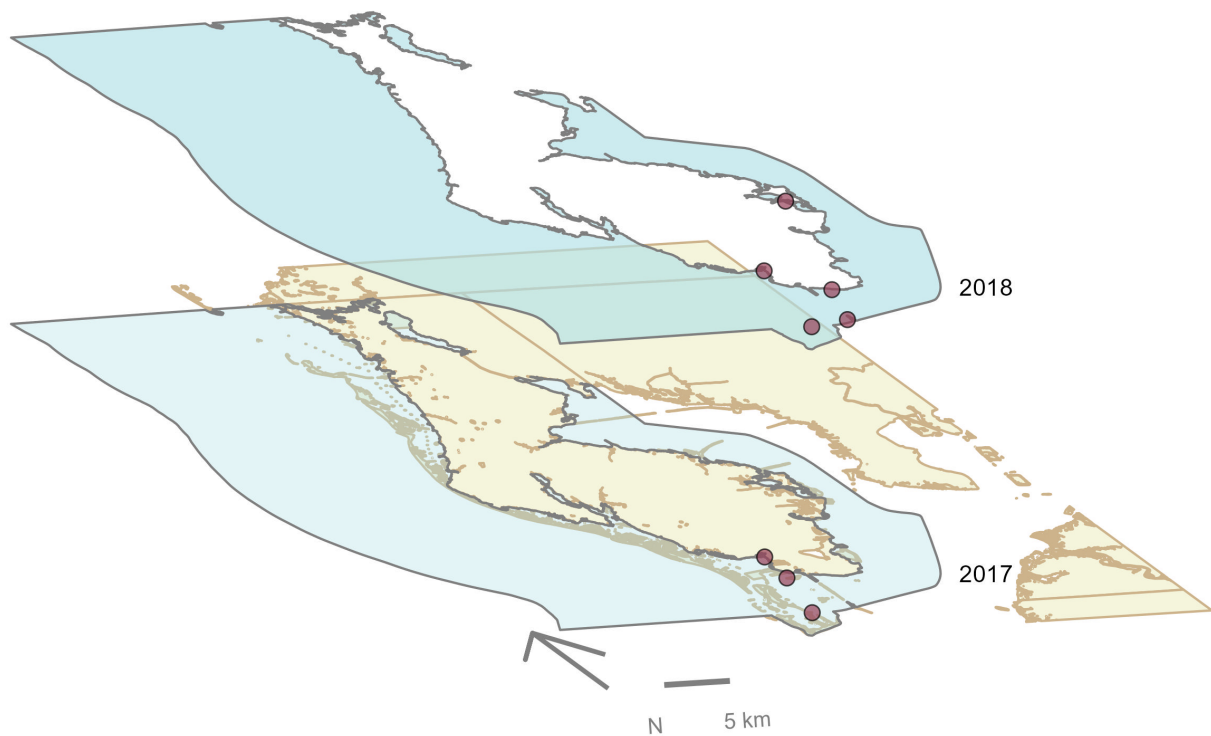
Figure 34: Map showing location of coastal wetlands sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

Pinellas County Aquatic Preserve

Oyster sample locations available in SEACAR by Program and Year



Program name

- Oyster Integrated Mapping and Monitoring Program (OIMMP) Pilot Monitoring Data

Figure 35: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Pinellas County Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Density

For natural reefs, there was insufficient data to calculate a trend for density.

Natural

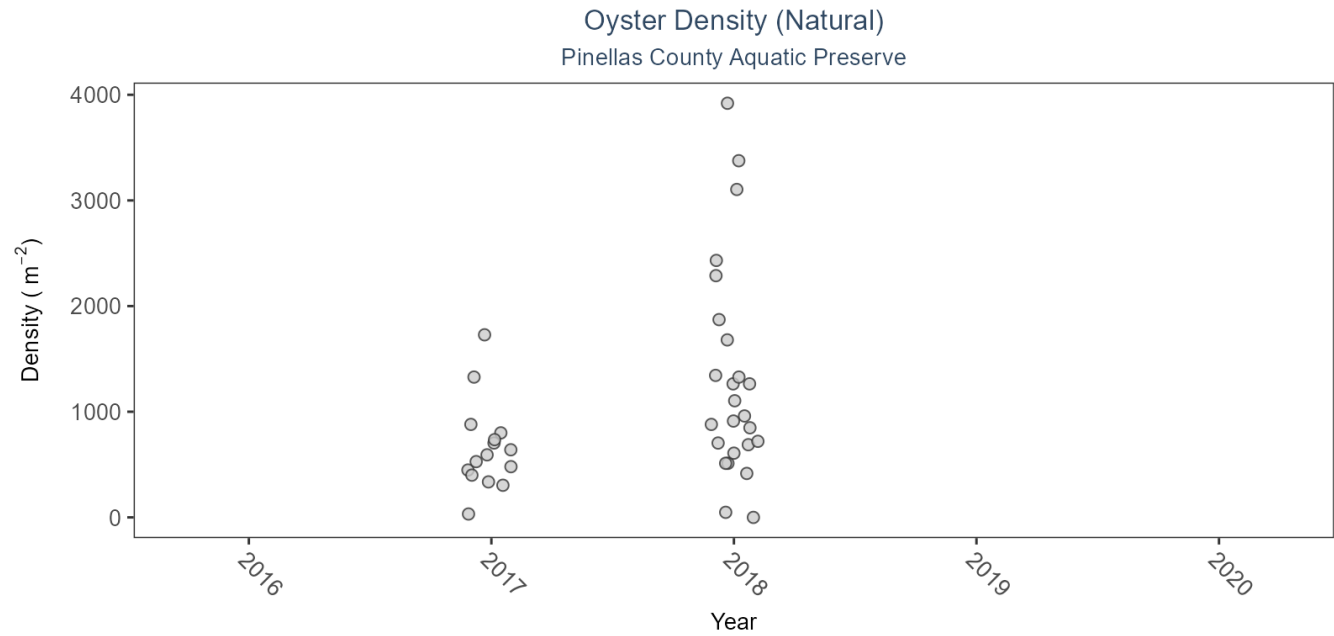


Figure 36: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 35: Model results for Oyster Density - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	Insufficient data	2017 - 2018	-	-	-

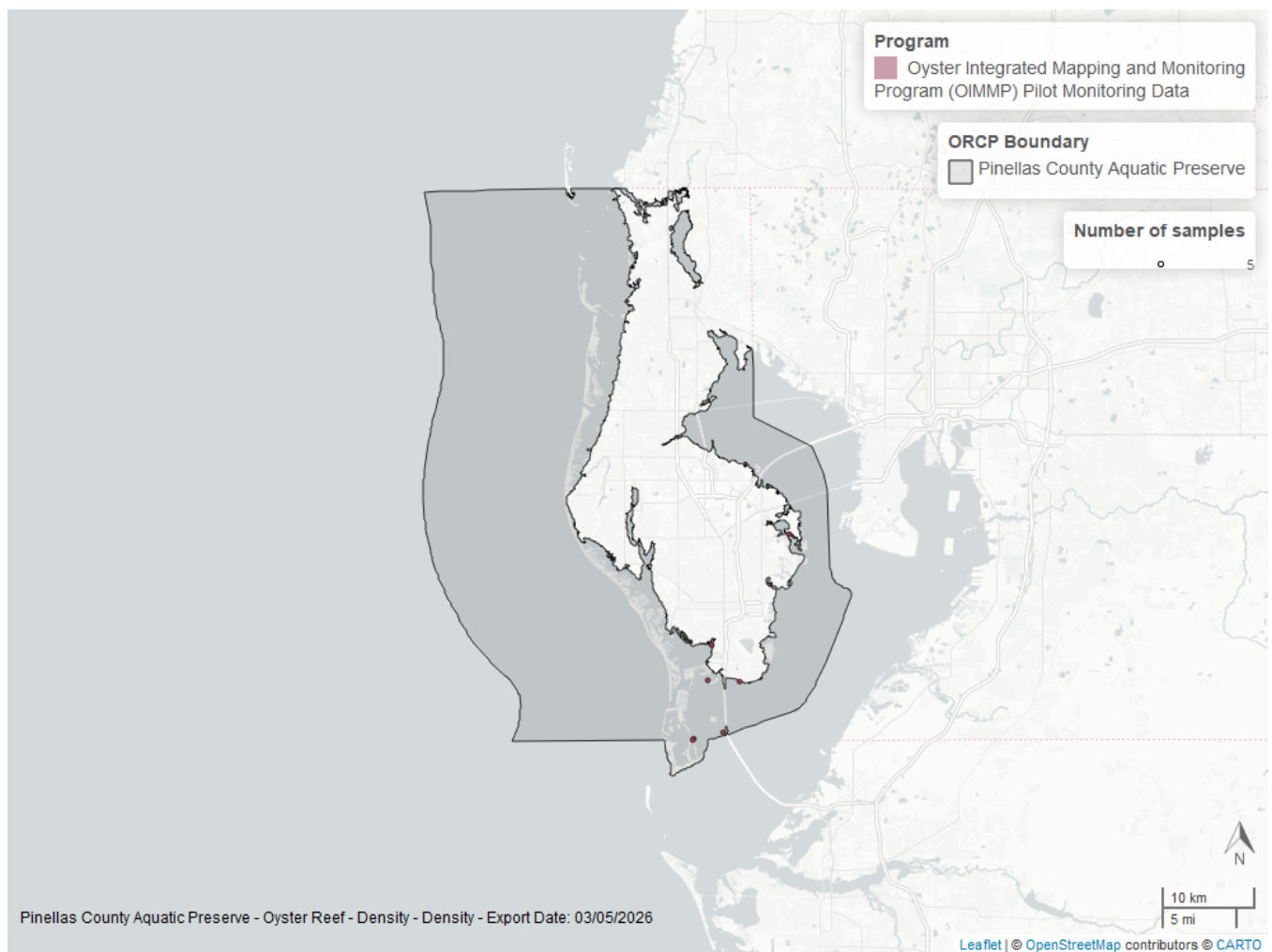


Figure 37: Map showing location of oyster density sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, there was insufficient data to calculate a trend for percent live cover.

Natural

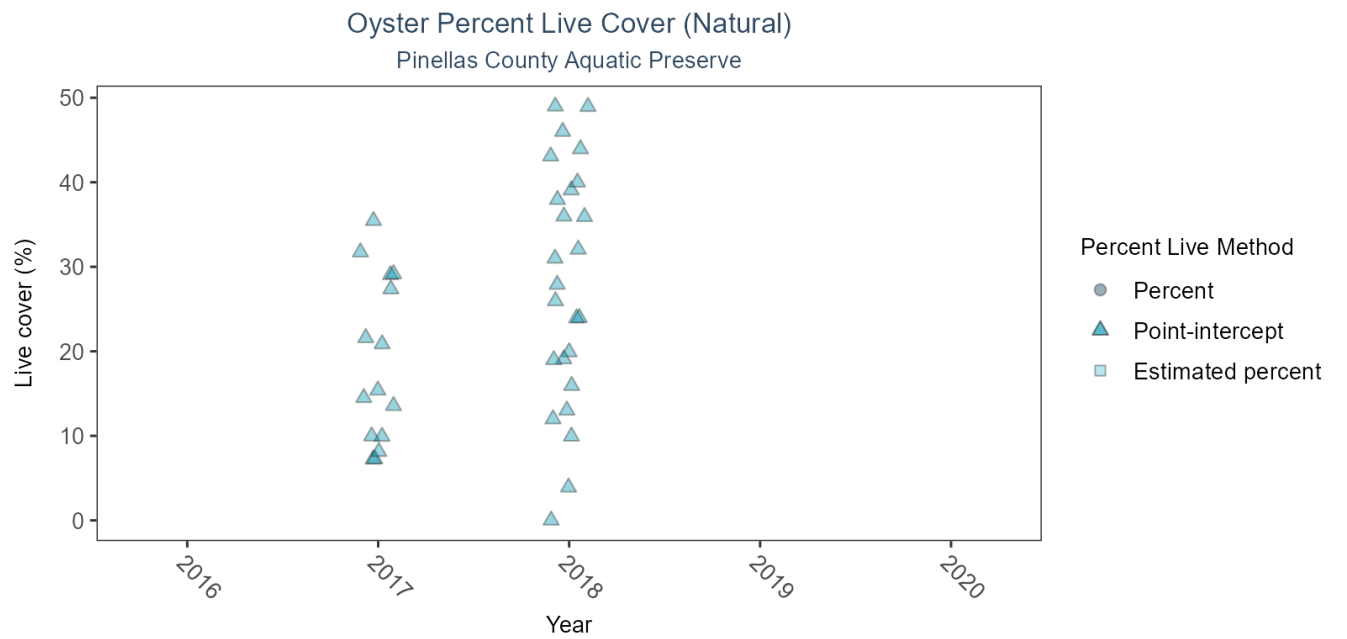


Figure 38: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 36: Model results for Oyster Percent Live - Natural

Habitat Type	Shell Type	Statistical Trend	Period of Record	Estimate	Standard Error	Credible Interval
Natural	Live Oysters	Insufficient data	2017 - 2018	-	-	-

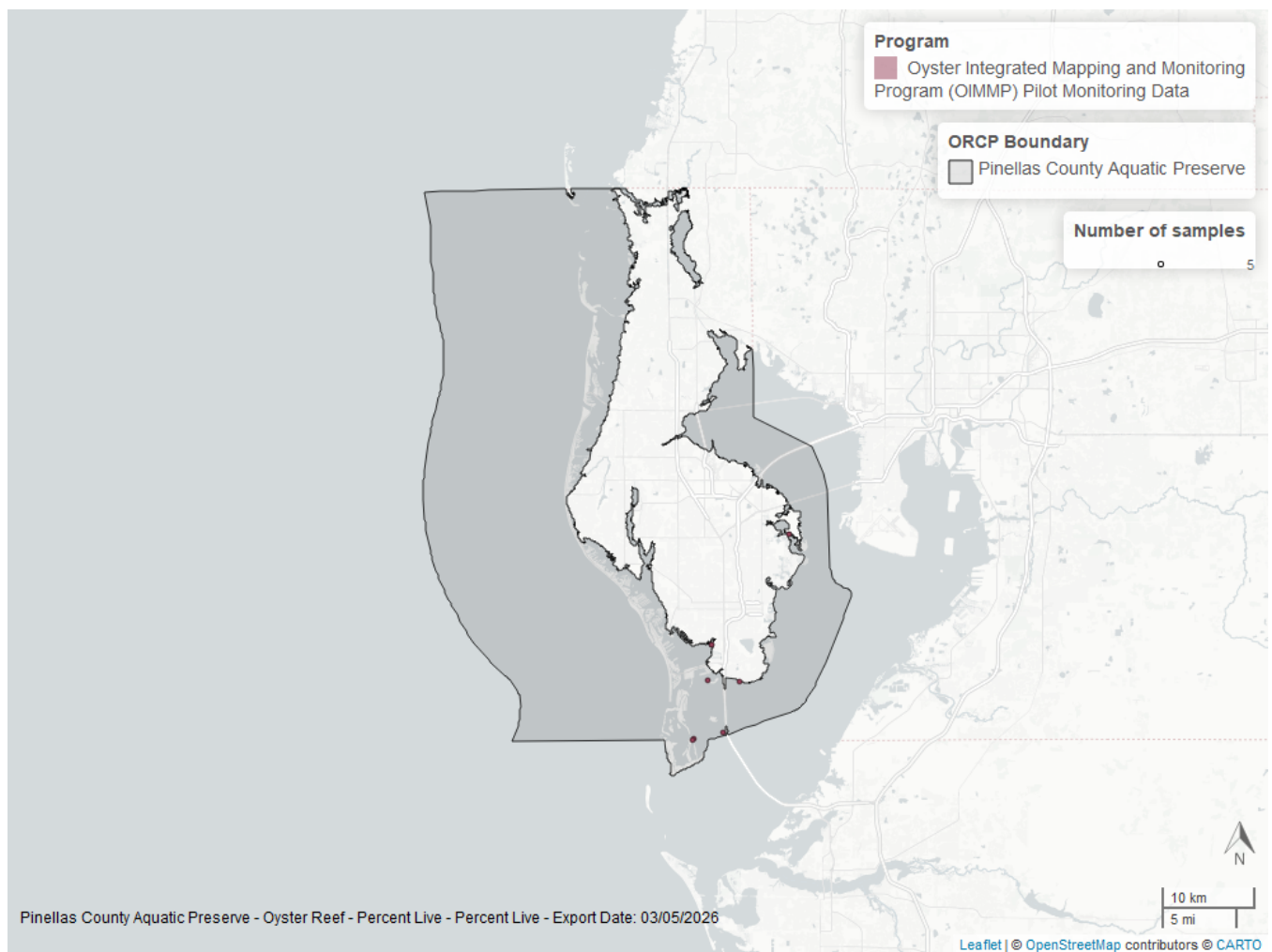


Figure 39: Map showing location of oyster percent live sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, there was insufficient data to calculate a trend for live oysters in either the 25-75mm or the ≥ 75 mm size class.

Natural

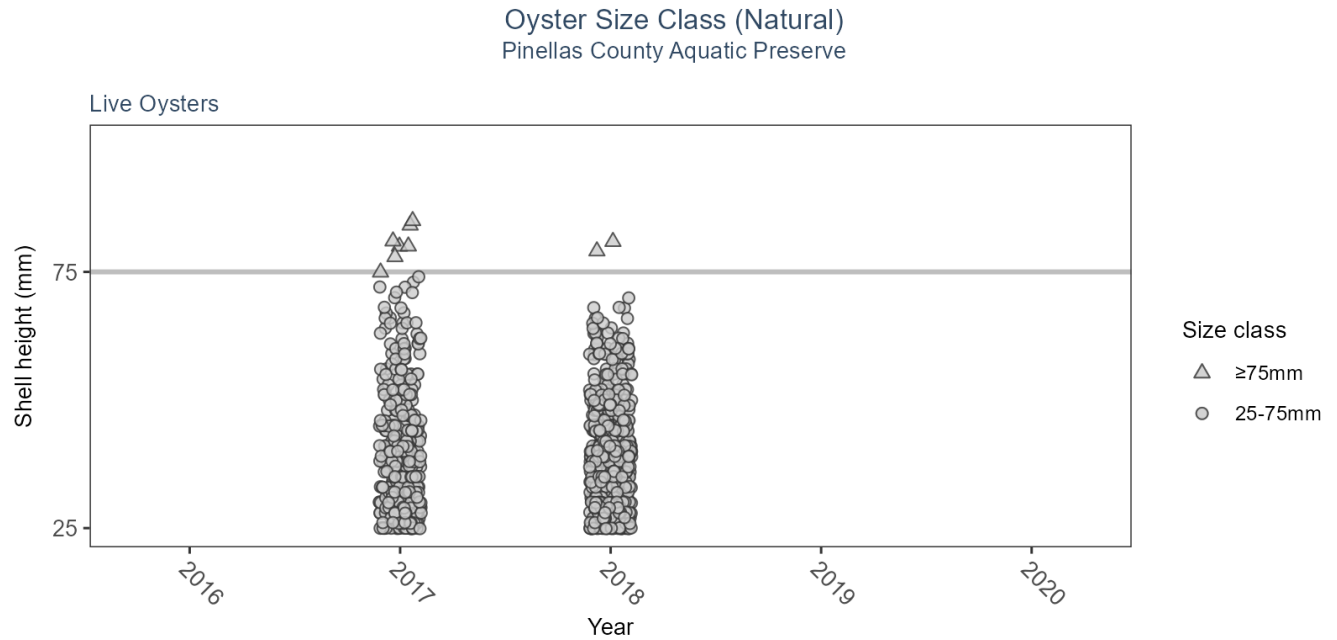


Table 37: Model results for Oyster Shell Height - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>SizeClass</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>No. of Samples</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	>75mm	Insufficient data	2017 - 2018	9	-	-	-
Natural	Live Oysters	25-75mm	Insufficient data	2017 - 2018	991	-	-	-

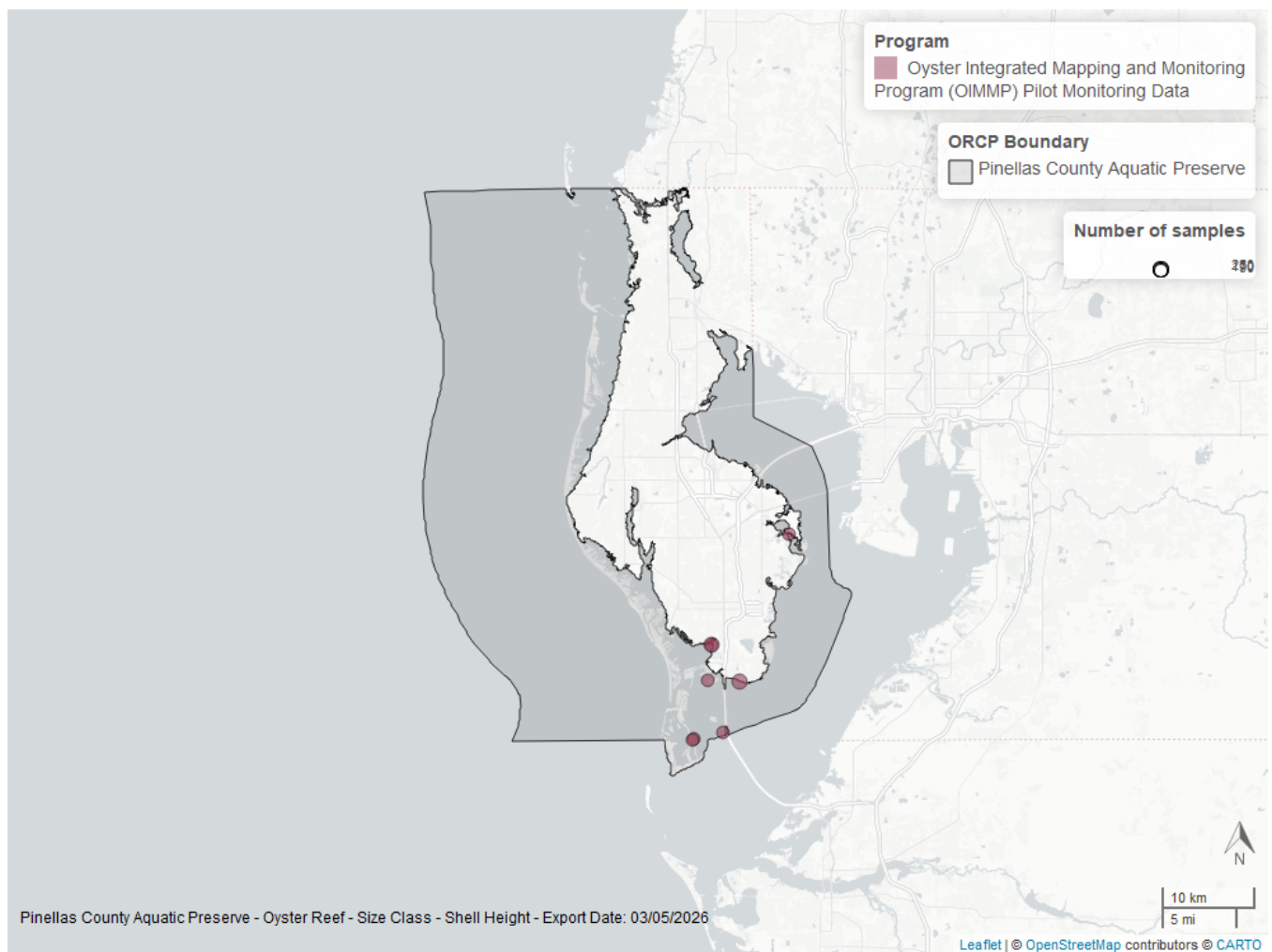


Figure 40: Map showing location of oyster shell height sampling locations within the boundaries of *Pinellas County Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

<i>Acanthophora</i> sp. ¹	Engraulidae spp. ³	<i>Ogcocephalus cubifrons</i> ³
<i>Acanthostracion quadricornis</i> ³	<i>Epinephelus itajara</i> ³	<i>Ogcocephalus</i> spp. ³
<i>Achirus lineatus</i> ³	<i>Epinephelus morio</i> ³	<i>Oligoplites saurus</i> ³
<i>Acrostichum danaeifolium</i>	<i>Etropus crossotus</i> ³	<i>Ophichthus gomesii</i> ³
<i>Aetobatus narinari</i> ³	<i>Eucinostomus argenteus</i> ³	<i>Ophidion grayi</i> ³
<i>Albula goreensis</i> ³	<i>Eucinostomus gula</i> ³	<i>Ophidion holbrookii</i> ³
<i>Albula</i> spp. ³	<i>Eucinostomus harengulus</i> ³	<i>Ophidion josephi</i> ³
<i>Albula vulpes</i> ³	<i>Eucinostomus</i> spp. ³	<i>Opisthonema oglinum</i> ³
Alpheidae spp. ³	<i>Eugerres plumieri</i> ³	<i>Opistognathus robinsi</i> ³
<i>Aluterus schoepfii</i> ³	<i>Eupatorium capillifolium</i>	<i>Opsanus beta</i> ³
<i>Aluterus scriptus</i> ³	<i>Ficus aurea</i>	<i>Oreochromis aureus</i> ³
<i>Anarchopterus criniger</i> ³	<i>Floridichthys carpio</i> ³	<i>Orthopristis chrysoptera</i> ³
<i>Anchoa cubana</i> ³	<i>Fundulus grandis</i> ³	Ostraciidae spp. ³
<i>Anchoa hepsetus</i> ³	<i>Fundulus similis</i> ³	Other green algae ¹
<i>Anchoa lyolepis</i> ³	<i>Gerres cinereus</i> ³	<i>Paraclinus fasciatus</i> ³
<i>Anchoa mitchilli</i> ³	<i>Gobiesox strumosus</i> ³	<i>Paraclinus marmoratus</i> ³
<i>Anchoa</i> spp. ³	Gobiidae spp. ³	<i>Paralichthys albigutta</i> ³
<i>Ancylosetta quadrocellata</i> ³	<i>Gobionellus oceanicus</i> ³	Penaeidae spp. ³
<i>Anguilla rostrata</i> ³	<i>Gobiosoma bosc</i> ³	<i>Penaeus duorarum</i> ³
<i>Anguilliformes</i> spp. ³	<i>Gobiosoma longipala</i> ³	<i>Penicillus</i> spp. ¹
<i>Archosargus probatocephalus</i> ³	<i>Gobiosoma robustum</i> ³	<i>Peprilus burti</i> ³
<i>Argopecten irradians</i>	<i>Gobiosoma</i> spp. ³	<i>Peprilus paru</i> ³
<i>Argopecten</i> spp.	<i>Gracilaria</i> sp. ¹	<i>Phoenix reclinata</i>
<i>Ariopsis felis</i> ³	<i>Gymnothorax saxicola</i> ³	<i>Pinus elliotii</i>
<i>Aster subulatus</i>	<i>Haemulon aurolineatum</i> ³	<i>Pluchea baccharis</i>
<i>Astrapogon alutus</i> ³	<i>Haemulon plumieri</i> ³	<i>Poecilia latipinna</i> ³
<i>Astroscoptes ygraecum</i> ³	<i>Haemulon</i> spp. ³	<i>Pogonias cromis</i> ³
Attached algae ¹	<i>Halichoeres bivittatus</i> ³	<i>Pomatomus saltatrix</i> ³
<i>Avicennia germinans</i> ²	<i>Halimeda incrassata</i> ¹	<i>Portunus</i> spp. ³
<i>Baccharis angustifolia</i>	<i>Halodule wrightii</i> ¹	<i>Prionotus rubio</i> ³
<i>Baccharis halimifolia</i>	<i>Halophila engelmannii</i> ¹	<i>Prionotus scitulus</i> ³
<i>Bacopa monnieri</i>	<i>Halophila</i> sp. ¹	<i>Prionotus</i> spp. ³
<i>Bagre marinus</i> ³	<i>Harengula jaguana</i> ³	<i>Prionotus tribulus</i> ³
<i>Bairdiella chrysoura</i> ³	<i>Hemicaranx amblyrhynchus</i> ³	<i>Pseudocrenilabrinae</i> ³
Balistidae spp. ³	<i>Hemiramphus brasiliensis</i> ³	<i>Quercus virginiana</i>
<i>Bascanichthys scuticaris</i> ³	<i>Hemiramphus</i> spp. ³	<i>Rachycentron canadum</i> ³
<i>Bathygobius soporator</i> ³	<i>Hippocampus erectus</i> ³	<i>Rhinoptera bonasus</i> ³
<i>Batis maritima</i> ²	<i>Hippocampus</i> spp. ³	<i>Rhizophora mangle</i> ²
<i>Bidens pilosa</i>	<i>Hippocampus zosterae</i> ³	<i>Rhizophytic algae</i> ¹
<i>Borrchia frutescens</i>	<i>Hypleurochilus caudovittatus</i> ³	<i>Rhizoprionodon terraenovae</i> ³
Bothidae spp. ³	<i>Hypleurochilus geminatus</i> ³	<i>Rimopenaeus constrictus</i> ³
<i>Brevoortia</i> spp. ³	<i>Hypnea</i> ¹	<i>Rimopenaeus</i> spp. ³
<i>Calamus arctifrons</i> ³	<i>Hyporhamphus meeki</i> ³	<i>Ruppia maritima</i> ¹
<i>Calamus penna</i> ³	<i>Hyporhamphus</i> spp. ³	<i>Sabal palmetto</i>
<i>Calamus proridens</i> ³	<i>Hyporhamphus unifasciatus</i> ³	<i>Salicornia bigelovii</i> ²
<i>Calamus</i> spp. ³	<i>Hypsoblennius hentz</i> ³	<i>Sardinella aurita</i> ³
<i>Callinectes ornatus</i> ³	<i>Ilex cassine</i>	<i>Sarotherodon melanotheron</i> ³
<i>Callinectes sapidus</i> ³	<i>Iva frutescens</i>	<i>Schinus terebinthifolia</i>
<i>Callinectes similis</i> ³	<i>Juncus roemerianus</i> ²	Sciaenidae spp. ³
<i>Callinectes</i> spp. ³	<i>Kyphosus sectatrix</i> ³	<i>Sciaenops ocellatus</i> ³
Carangidae spp. ³	<i>Kyphosus</i> spp. ³	<i>Scomberomorus maculatus</i> ³
<i>Caranx bartholomaei</i> ³	<i>Lachnolaimus maximus</i> ³	<i>Scorpaena brasiliensis</i> ³
<i>Caranx crysos</i> ³	<i>Lactophrys trigonus</i> ³	<i>Selene vomer</i> ³
<i>Caranx hippos</i> ³	<i>Lagodon rhomboides</i> ³	<i>Serenoa repens</i>

Caranx latus ³	Laguncularia racemosa ²	Serraniculus pumilio ³
Caranx spp. ³	Leiostomus xanthurus ³	Serranidae spp. ³
Carcharhinus acronotus ³	Lepisosteus osseus ³	Serranus subligarius ³
Carcharhinus leucas ³	Limulus polyphemus	Sesuvium portulacastrum ²
Carcharhinus limbatus ³	Lobotes surinamensis ³	Sicyonia brevirostris ³
Caulerpa ¹	Lophogobius cyprinoides ³	Sicyonia laevigata ³
Caulerpa mexicana ¹	Lucania parva ³	Sicyonia spp. ³
Caulerpa prolifera ¹	Lutjanus analis ³	Sicyonia typica ³
Caulerpa sertularioides ¹	Lutjanus apodus ³	Smilax auriculata
Caulerpa spp. ¹	Lutjanus griseus ³	Solidago sempervirens
Centropomus undecimalis ³	Lutjanus spp. ³	Spartina bakeri ²
Centropristis philadelphica ³	Lutjanus synagris ³	Spartina patens ²
Centropristis striata ³	Lycium carolinianum	Sphoeroides nephelus ³
Chaetodipterus faber ³	Lyngbya sp.	Sphoeroides spengleri ³
Chaetodon capistratus ³	Malaclemys terrapin ³	Sphyraena barracuda ³
Chaetodon ocellatus ³	Maytenus phyllanthoides	Sphyraena borealis ³
Chasmodes saburrae ³	Menidia spp. ³	Sphyraena guachancho ³
Chelonia mydas ³	Menippe mercenaria ³	Sphyrna tiburo ³
Chilomycterus schoepfii ³	Menippe spp. ³	Sporobolus virginicus ²
Chloroscombrus chrysurus ³	Menticirrhus americanus ³	Stephanolepis setifer ³
Citharichthys macrops ³	Menticirrhus littoralis ³	Strongylura marina ³
Cladium mariscus	Menticirrhus saxatilis ³	Strongylura notata ³
Conocarpus erectus ²	Menticirrhus spp. ³	Strongylura timucu ³
Cyanobacteria	Microgobius gulosus ³	Syacium papillosum ³
Cynoscion arenarius ³	Microgobius spp. ³	Symphurus plagiusa ³
Cynoscion nebulosus ³	Microgobius thalassinus ³	Syngnathus floridae ³
Cyperus odoratus	Micropogonias undulatus ³	Syngnathus louisianae ³
Cyprinodon variegatus ³	Mikania scandens	Syngnathus scovelli ³
Diapterus auratus ³	Monacanthus ciliatus ³	Syngnathus springeri ³
Digenea simplex ¹	Monacanthus spp. ³	Synodus foetens ³
Diodon holocanthus ³	Mugil cephalus ³	Syringodium filiforme ¹
Diplectrum formosum ³	Mugil curema ³	Taxodium distichum
Diplectrum spp. ³	Mugil spp. ³	Telmatoblechnum serrulatum
Diplodus holbrookii ³	Mugil trichodon ³	Thalassia testudinum ¹
Distichlis littoralis ²	Mycteroperca bonaci ³	Total SAV ¹
Dorosoma cepedianum ³	Mycteroperca microlepis ³	Toxicodendron radicans
Dorosoma petenense ³	Myrica cerifera	Trachinotus carolinus ³
Drift algae ¹	Myrophis punctatus ³	Trachinotus falcatus ³
Drift red algae ¹	Narcine bancroftii ³	Trinectes maculatus ³
Echeneis naucrates ³	Negaprion brevirostris ³	Tylosurus crocodilus ³
Echeneis neucratoides ³	Nicholsina usta ³	Unidentified species
Echeneis spp. ³	No fish	Urophycis floridana ³
Eleocharis interstincta	No gear set	Vaucheria ¹
Elopiformes spp. ³	No grass in quadrat ¹	Vigna luteola
Elops saurus ³	Ocyurus chrysurus ³	Acanthophora sp. ¹

1 - Submerged Aquatic Vegetation, 2 - Coastal Wetlands, 3 - Nekton

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